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Computer Science Research Report

MetaShift-Bench: A Target-Fixed Benchmark for Selective Scope Answerability

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Abstract

A time-series change is not automatically a scope conclusion. A target-only observation can reveal that a constructed target path changed, but cannot by itself determine whether the change is local to that target or shared with a comparison network. This report introduces MetaShift-Bench as an information-constrained selective scope-auditing benchmark. It holds the target path exactly fixed across paired local and shared synthetic worlds, varies only the donor-side scope information, and asks which fraction of scope questions a frozen policy may answer at a stated error tolerance. The contribution is an evaluation problem and evidence architecture, not an estimator-superiority or physical-monitoring diagnosis claim.

The preregistered v0.5 experiment uses 360 independent held-out synthetic components, 230,400 matched pairs, and a complete 640-cell design crossing donor participation, signal strength, mismatch, availability, contamination, nonlinear raw-scale leakage, and bounded noise. For the finite predeclared policy set, the target-only answerability envelope is zero at all four reported tolerances ($\alpha \in \{.01, .05, .10, .20\}$), while the comparative envelope is also zero through $\alpha = .10$ and reaches 0.603746 at $\alpha = .20$ with observed conditional error 0.195208. A structural interval-separation certificate answers 0.390612 of matched pairs with 0.000000 observed conditional error and 0.976583 efficiency relative to a predeclared simulation-information oracle region. It abstains on 140,403 pairs with nonpositive structural margin, including all 46,080 observationally identical $q = 0$ negative controls. Zero observed error is not a population-risk guarantee, and the certificate uses synthetic design information unavailable in deployment.

The report proves the target-only impossibility statement under matched balanced target laws, specializes information-refinement monotonicity, derives the exact availability-normalized partial-scope residual gap, and gives a bounded-error sufficient condition for structural abstention. The older v0.4 construction is retained only as an endpoint sanity check; the frozen v0.3.2 AQS audit remains deployment and abstention context, not scope-label accuracy evidence. Its negative MetaShift-versus-standard-synthetic-control result is preserved: it does not support a general MetaShift superiority claim. No taxonomy-stratified analysis is reported, and no result identifies a real measurement mechanism, device state, causal bias, or corrected concentration.

Keywords: selective classification; information-constrained inference; scope auditing; abstention; synthetic evaluation; reproducibility.

Contents

1 Introduction

An observed time-series break and a defensible explanation of its scope are different computational objects. A target path may plainly change while the available observation contains no information about whether that change is local, shared across a network, or associated with a physical mechanism. A numerical classifier can still emit a label in that setting, but its confidence does not create missing information. The central question of this report is therefore not "which estimator wins?" It is:

> *Given a specified observation channel and an allowed scope-error rate, what fraction of scope questions can a system answer, and when must it abstain?*

We call this *information-constrained selective scope auditing*. It separates three questions that are frequently conflated:

1. **Detection:** did the target path change?
2. **Scope:** is the target change local or shared with a declared comparison network?
3. **Mechanism:** is a measurement, device, maintenance, processing, or other physical process responsible?

Detection can be studied from a target series. Scope requires valid comparative information. Mechanism additionally requires records, technical metadata, and human review. The last distinction is especially important for public air-quality data: a reported Method Code transition is an observable metadata anchor, not evidence that a device was replaced, failed, or biased.

MetaShift-Bench supplies a deliberately target-fixed synthetic test of the second question. Every matched pair has byte-identical target observations in the local and shared arms. Only donor participation changes. Consequently, a target-only policy cannot distinguish balanced local from shared labels in the stipulated experiment, whereas a policy that receives the target and donor channel can potentially do so. The benchmark measures a finite-policy *Scope Answerability Frontier*: maximum held-out answer coverage among predeclared policies whose observed answered-case scope error is at most a specified tolerance. The difference between comparative and target-only frontiers is named the *Scope Answerability Gain*.

The central v0.5 design extends the earlier binary local-versus-regional endpoint into a donor-participation continuum. It crosses nominal participation $q \in \{0, .25, .50, .75, 1\}$ with signal strength and five adverse comparison conditions. A bounded structural certificate answers only when its availability-normalized score intervals are provably separated under the generator's declared envelopes. This makes an unfavorable outcome useful: nonpositive structural margin is reported as a required abstention region, not hidden as a weak classification score.

The report asks five linked questions:

1. **RQ0:** What answerability frontier is achieved by target-only, comparative, and synthetic-design-assisted policies across prespecified scope-error tolerances?
2. **RQ1:** Under which assumptions is target-only scope inference impossible, and how does donor participation create exact comparative separation on the stated analysis scale?
3. **RQ2:** Does a structural certificate safely abstain when the declared comparison channel cannot separate local from partially shared scope?
4. **RQ3:** What does the earlier v0.4 endpoint check establish, and what does it fail to establish about a continuous boundary?
5. **RQ4:** What role remains for the v0.3.2 AQS benchmark and the negative MetaShift-versus-standard-synthetic-control comparison?

Our bounded contribution is fourfold. First, we instantiate a target-fixed paired benchmark that measures selective scope answerability by information channel, rather than treating scope accuracy as an unconditional classifier score. Second, we formalize a partial-scope construction with availability-normalized donor participation and derive its exact residual separation for the defined mean score. Third, we test a predeclared bounded structural abstention condition on a full adverse-condition grid, preserving all failures and $q = 0$ negative controls. Fourth, we retain an independently frozen public-data audit as deployment context while refusing to convert its metadata anchors into scope labels or physical mechanisms.

This is not a claim to invent selective classification, risk–coverage analysis, comparison of statistical experiments, common-versus-idiosyncratic change detection, fault distinguishability, or synthetic control. The related work and theory delimit what is established from what is specialized to this synthetic setting. No taxonomy-stratified analysis is reported because the Method Code taxonomy remains pending independent human review.

2 Problem formulation and claim boundaries

2.1 Three questions and observation channels

Let $Y \in \{L, S\}$ denote a stipulated synthetic scope label, where L is target-local and S is shared with comparative donors. Let X be the declared target path; let D contain donor paths, fixed donor weights, and availability indicators; and let E denote external records when such records are admissible. The conceptual nested channels are

$$O_T = X, \qquad O_C = (X, D), \qquad O_E = (X, D, E).$$

The scope question is distinct from detection and mechanism. Detection asks whether a target changed. Scope asks whether the target change is local or shared under a declared comparative construction. Mechanism asks why it occurred. The v0.5 experiment supplies constructed labels only for the first two questions. It contains no admissible external-evidence channel or mechanism labels, so it does not empirically estimate an O_E frontier.

A selective policy $\phi = (g, s)$ maps an allowed observation channel and auxiliary randomness to a scope prediction g and an answer indicator $s \in \{0, 1\}$. Its population answerability frontier at tolerated conditional scope error α is

$$\Gamma_{\mathcal{I}}(\alpha) = \sup_{\substack{\phi\ \mathcal{I}\text{-measurable} \\ \Pr(s_\phi=1)>0 \\ \Pr(g_\phi(O)\neq Y|s_\phi=1)\leq\alpha}} \Pr(s_\phi = 1).$$

The supremum is defined as zero if no positive-coverage policy meets the constraint. This is an inverse selective risk–coverage objective specialized to a scope label; it is not a new generic risk–coverage definition [22, 23].

For nested channels, the *Scope Answerability Gain* is

$$\mathrm{SAG}_{\mathcal{I}\to\mathcal{J}}(\alpha) = \Gamma_{\mathcal{J}}(\alpha) - \Gamma_{\mathcal{I}}(\alpha).$$

The benchmark-specific name does not make its nonnegativity a new decision-theoretic principle. If the richer channel can discard its additional coordinates, its policy set includes the poorer-channel policy set, a Blackwell-style information-refinement argument [14, 15].

2.2 Target-only impossibility and finite-policy estimation

Proposition 1 (target-only impossibility). Suppose the synthetic labels are balanced and the matched construction satisfies

$$\Pr(Y = L) = \Pr(Y = S) = \frac{1}{2}, \qquad \mathcal{L}(X \mid Y = L) = \mathcal{L}(X \mid Y = S).$$

If selection and prediction use only X and auxiliary randomness that is conditionally independent of Y given X, then every positive-coverage target-only policy has conditional scope error $1/2$. Therefore

$$\Gamma_T(\alpha) = 0 \qquad \text{for } \alpha < \frac{1}{2}.$$

Proof. Conditioning on a target-only answer does not change the balanced posterior label probability because the accepted target law is the same under both labels. Either binary prediction is consequently wrong with probability $1/2$. □

This statement would fail if target-only input leaked a pair identifier, row order, donor availability summary, participation value, label schedule, generator truth, or label-dependent auxiliary randomness. The v0.5 execution contract forbids those side channels and tests target identity explicitly.

Proposition 2 (nested-channel monotonicity). When O_C contains O_T under the same joint law, label space, loss, and policy class,

$$\Gamma_C(\alpha) \geq \Gamma_T(\alpha).$$

The result follows by implementing every target-only policy after discarding D. It neither implies strict improvement nor validates any finite-sample comparative classifier.

The frozen execution estimates neither unrestricted population frontier. Instead it reports the held-out finite-policy envelope

$$\widehat{\Gamma}^{\mathcal{P}}_{\mathcal{I}}(\alpha) = \max_{\substack{\phi \in \mathcal{P} \\ \widehat{R}_\phi \leq \alpha}} \widehat{C}_\phi,$$

with zero when no positive-coverage frozen policy qualifies. Here $\mathcal{P}$ contains the target-only forced rule, comparative forced rule, four separately calibration-selected confidence rules, and the specified synthetic-design-assisted certificate. Thus every displayed result is a descriptive held-out envelope for the frozen generator and policy set, not a population optimum, conditional-risk guarantee, or post-evaluation policy choice.

2.3 Target-fixed partial-scope construction

The exact v0.5 construction is performed on the analysis scale $z = f(y) = \log(1 + \max(y, 0))$. Let $w_j \geq 0$ be fixed donor weights and a_{jt} denote donor availability. On every retained date,

$$\widetilde{w}_{jt} = \frac{w_j a_{jt}}{\sum_k w_k a_{kt}}, \qquad q_t = \sum_j \widetilde{w}_{jt} \lambda_{jt},$$

where $\lambda_{jt} \in [0, 1]$ is declared donor participation. For positive post-anchor schedule h_t, the paired worlds are

$$z^L_t = z_t + h_t, \quad z^L_{jt} = z_{jt}, \qquad z^S_t = z_t + h_t, \quad z^S_{jt} = z_{jt} + \lambda_{jt} h_t.$$

The target path is exactly identical in both worlds. With fixed offset b,

$$r_t = z_t - \sum_j \widetilde{w}_{jt} z_{jt} - b, \qquad A = \mathrm{mean}_{W_{\mathrm{post}}}\, r_t - \mathrm{mean}_{W_{\mathrm{pre}}}\, r_t.$$

Proposition 3 (construction-specific residual identity). With arm-invariant availability, retained dates, weights, preprocessing, and pre-anchor offset,

$$r_t^L - r_t = h_t, \qquad r_t^S - r_t = (1 - q_t)h_t.$$

For a constant post schedule $h_t = H$, the score-center separation is

$$A^L - A^S = \overline{q_t H}.$$

For a variable schedule the correct expression is $\overline{q_t h_t}$, not $\bar{q}\,\bar{h}$. This is exact algebra for the stipulated analysis-scale construction, not a generic identifiability claim. A raw-scale additive field does not preserve this identity after the nonlinear transformation and is treated only through a conservative leakage envelope.

2.4 Structural Answerability Certificate

The certificate is deliberately a synthetic-design-information-assisted policy, denoted here by O_C^+ rather than by the unobserved real-world O_E. It receives the generated participation and bounded-error quantities that a deployed monitoring system would not know. Suppose its local and shared scores obey

$$|A^L - H| \le B_L, \qquad |A^S - (H - G)| \le B_S,$$

and all scored retained dates satisfy $q_t \ge q_{\min}$ and $h_t \ge H_{\min} > 0$. Let

$$G_{\min} = q_{\min} H_{\min}, \qquad \kappa = G_{\min} - (B_L + B_S).$$

Proposition 4 (interval-separation certificate). If $\kappa > 0$, the local and shared score intervals are disjoint. Every threshold in

$$(H - G_{\min} + B_S,\ H - B_L)$$

separates them under the stated bounds. The implementation uses their midpoint,

$$\tau_{\text{cert}} = H - \frac{G_{\min}}{2} + \frac{B_S - B_L}{2},$$

answers only if $\kappa > 0$, and predicts local if $A > \tau_{\text{cert}}$.

Proof. The upper shared bound is $H - G_{\min} + B_S$ and the lower local bound is $H - B_L$; their strict ordering is exactly $\kappa > 0$. □

The theorem requires bounded noise, fixed retained-date semantics, valid availability normalization, modeled donor mismatch, bounded contamination, and a valid clipping-aware raw-field bound. Nonfinite values, changed dates, unmodeled transformations, fewer than three donors, or an envelope violation invalidate the certificate and require abstention. A nominal midpoint $H - G_{\min}/2$ is not generally safe under asymmetric bounds; the asymmetric term above is necessary. The certificate does not establish a physical sensor fault, a real-network policy, or a confidence-based proof of identifiability.

2.5 AQS deployment context and observation boundaries

The v0.3.2 public-data audit provides a separate deployment context. Let $s = (\text{state}, \text{county}, \text{site})$ be a physical monitoring site, p a reported POC, and $y_{s,p,t}$ a retained daily concentration. An AQS anchor is

$$a = (s, p, t_0, m^-, m^+),$$

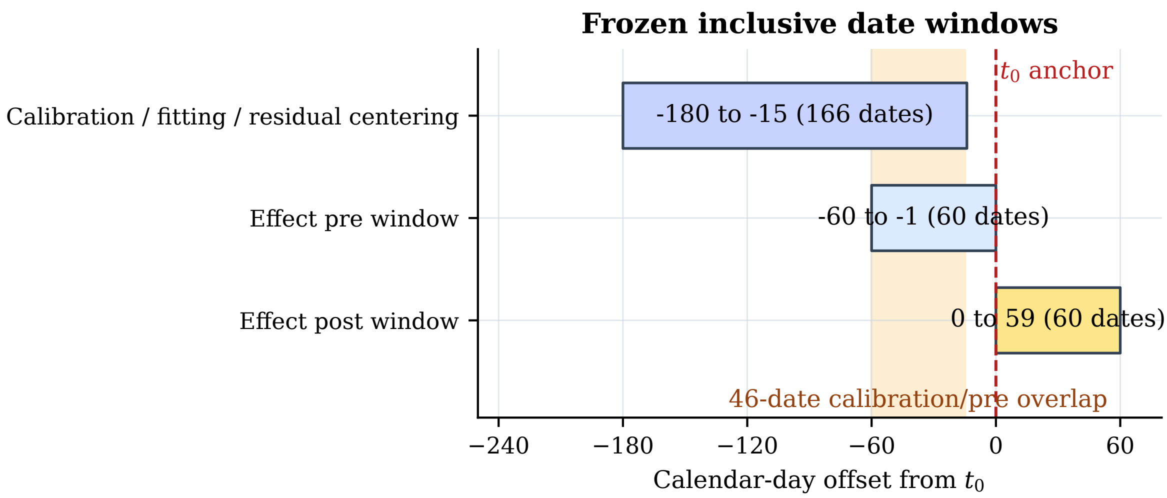

Figure 1: Inclusive date-window implementation used by the frozen v0.3.2 deployment audit. The calibration and displayed pre-anchor slices overlap on 46 calendar dates. No post-anchor outcome is fitted, but the overlap prevents treating displayed pre-anchor residuals as an independent validation set.

Table 1: Claim boundaries imposed by the MetaShift-Bench protocol.

The evidence can support	The evidence does not establish	Required interpretation
A reproducible metadata-anchored local residual audit	That a Method Code transition is a physical replacement or fault	Treat the transition as an anchor, not physical ground truth.
Benchmark-specific synthetic tradeoffs among frozen methods	A robust aggregate MetaShift superiority claim	Report the paired intervals and retain the negative result.
Protocol-defined evidence tiers and abstentions	Instrument bias, true pollution correction, or causal attribution	Use tiers only to prioritize further records-based review.
Conditional interval behavior on frozen synthetic data	Coverage-calibrated confidence intervals for real-event physical bias	Describe real-event intervals as diagnostic.

Note. These boundaries are design constraints, not post hoc caveats.

where t_0 is a persistent reported Method Code transition. This anchor is observable metadata, not a verified intervention. The deployment audit asks whether a pre-specified residual can be computed and whether its observational diagnostics are available; it does not provide synthetic scope labels or physical mechanism truth.

For a qualified real target $i = (s, p)$ and donors $j \in N_a$, the frozen deployment residual is

$$r_{i,t} = z_{i,t} - \sum_{j \in N_a} w_{ij} z_{j,t} - \mathrm{median}_{u \in T_{\mathrm{cal}}} \left(z_{i,u} - \sum_{j \in N_a} w_{ij} z_{j,u} \right).$$

Its displayed effect is the post-minus-pre median residual. It remains a diagnostic local residual discontinuity on the log scale, not a causal physical-bias estimate or corrected concentration.

2.6 Claim boundaries

The public-data audit records metadata anchors, synthetic evaluations, and availability failures without promoting weak records to mechanism labels [24, 36]. The answerability results specialize only to a new bounded synthetic generator. The method taxonomy remains human-blocked: no taxonomy-stratified analysis is reported.

3 Background and related work

3.1 Information, selective prediction, and answerability

Blackwell's comparison of statistical experiments formalizes when one observation can reproduce another decision problem by discarding or garbling information [14, 15]. The nested-channel monotonicity in this report is a direct policy-set specialization of that established idea. It does not imply that a fitted rule benefits from more features, that a finite sample achieves the population frontier, or that comparative records are trustworthy in a real monitoring network.

Selective classification and reject-option learning already study the risk–coverage trade-off, optimal coverage subject to a desired risk, and the cost of abstention [10, 18, 22, 23, 27]. Hierarchical selective classification further permits a less-specific answer when a fine label is not justified [29]. These are important precedents, not contributions claimed here. MetaShift-Bench specializes the object being selected: the answer is a synthetic local-versus-shared *scope* judgment under explicitly contrasting information channels. The held-out envelope is deliberately descriptive. Calibration-selected confidence does not itself provide a distribution-free conditional-risk guarantee [7].

3.2 Common, idiosyncratic, and weakly labelled changes

Multivariate change-point work distinguishes common and idiosyncratic components or tests common breaks in panels [5, 8]. Those methods motivate the scope distinction, but they do not make an availability-normalized donor-participation continuum or this benchmark's target-fixed paired design standard. Conversely, weak-supervision research studies inference with partial or proxy labels [17, 38]. The v0.5 scope labels are generator-defined gold labels, whereas a real AQS Method Code remains only a weak metadata anchor. Neither form of evidence identifies a physical mechanism.

3.3 Residual signatures, fault diagnosis, and sensor placement

Analytical redundancy, residual signatures, fault distinguishability, and sensor placement are established in model-based diagnosis [19, 33, 40]. They require physical relations, fault models, or sensor-network assumptions that this project does not possess. The structural certificate is therefore not claimed as a new fault-diagnosis method, a sensor-observability result, or a real-device isolation procedure. It is a bounded sufficient condition for abstention in the stipulated synthetic score construction.

3.4 Monitoring comparability and public metadata

Monitoring comparability is an empirical property rather than a label inferred from one administrative field. EPA publishes AQS file formats, Method Code and POC references, and continuous-monitor comparability resources [43, 44, 46, 47]. Collocated Federal Reference Method and Federal Equivalent Method studies show why configuration-specific comparison matters [31]. They do not, however, provide dated physical-change ground truth for every historical AQS Method Code transition.

The climate-record homogenization literature offers a useful structural analogy: pairwise comparisons and station histories can expose inhomogeneities, while the cause remains conditional on the available documentation [37]. Air-pollution analyses similarly use meteorological normalization to distinguish interventions or trends from changing weather [26, 30, 51]. Low-cost PM2.5 sensor studies further motivate calibration and reference comparison [9, 20, 21].

Their instruments, target populations, and calibration goals differ from a retrospective audit of regulatory-monitor metadata. MetaShift-Bench uses this literature as measurement-quality context, not as evidence that a particular AQS transition has a particular physical cause.

3.5 Reference series, counterfactuals, and change points

Single-series change-point procedures include cumulative-sum monitoring, segmentation methods such as PELT, and multiscale approaches [25, 32, 41]. These methods are valuable baselines because they find patterns in one sequence; they cannot by themselves determine whether a break is monitor-local or shared across a region. The difference motivates matched regional variants in this benchmark.

Synthetic control provides donor-weighted counterfactuals and in-space placebo logic [2, 3]. Generalized, augmented, and synthetic difference-in-differences extensions address other panel structures and regularization choices [4, 11, 50]. The modern synthetic-control overview emphasizes that data requirements and failure conditions remain substantive [1]. Modern difference-in-differences work also emphasizes assumptions and heterogeneity under varied treatment timing [16, 28]. MetaShift-Bench adopts the computational ideas of pre-event fitting and reference comparison, but does not claim that the causal identification assumptions of a policy study apply to a reported monitoring-method transition.

3.6 Uncertainty, selection, and abstention

Time-series resampling must respect serial dependence; circular moving-block bootstrap methods are a natural descriptive tool for this purpose [34]. Distribution-free predictive and conformal methods offer another way to assess interval behavior when exchangeability assumptions are made explicit [6, 35, 39, 48]. In this report, these methods motivate synthetic coverage evaluation rather than a claim that real monitoring events enjoy generic calibration.

The evidence-tier task is also related to multiple testing, selection, and abstention. Benjamini–Hochberg adjustment organizes an exploratory family of screening quantities [12]; it does not turn those quantities into causal probabilities. Post-selection inference highlights why selection can invalidate naive uncertainty statements [13]. Research on noisy labels underscores that observed proxies are not ground truth [24, 36]. MetaShift-Bench consequently records an inconclusive tier when required comparative evidence is unavailable instead of forcing a positive or negative physical diagnosis.

3.7 Contribution boundary

Table 2 locates the benchmark relative to its component literatures. The comparison concerns the scope of the reported workflow, not a claim that MetaShift-Bench improves the underlying methods.

The bounded joint contribution is a target-fixed donor-participation continuum that evaluates channel-specific selective scope answerability and a bounded-error structural abstention condition in a synthetic audit setting. It combines established ingredients rather than claiming their invention: information refinement, selective risk–coverage, common/idiosyncratic changes, donor-weighted comparisons, pairwise monitoring comparisons, and residual diagnosis are all prior work. Pairwise homogenization in particular has long used reference comparisons and controlled simulations [37, 49]. We did not identify, in a focused primary-source audit, a source stating this complete joint construction. That absence is not proof of priority. The report makes no "first ever" claim, no generic risk–coverage claim, no estimator-superiority claim, and no real

Table 2: Relationship between MetaShift-Bench and contributing research traditions.

Research tradition	Typical object and evidence	MetaShift-Bench boundary
Synthetic control and DiD [2, 4]	Pre-intervention panel outcomes and comparison units for an estimand under explicit assumptions.	Uses pre-anchor donor fitting and placebos as diagnostics; does not assert policy-style causal identification for metadata transitions.
Single-series change points [32, 41]	One sequence and a change statistic or estimated break.	Retains these methods as baselines but separates a target-local pattern from a matched regional movement only when a qualified reference is available.
Monitoring comparability and collocation [31, 46]	Configuration-specific technical comparisons, collocation, and station documentation.	Treats auxiliary POC, QA, and document records as contextual evidence; no metadata field is promoted to physical-device ground truth.
Climate-record homogenization [37]	Station histories plus pairwise evidence for record inhomogeneities.	Adopts the audit logic of preserving candidate discontinuities and requiring records for mechanism; it does not claim a homogenized corrected PM2.5 series.

monitoring-mechanism identification claim.

4 Data and benchmark construction

4.1 v0.5 synthetic input boundary

The primary scope-answerability experiment does not use an observational corpus. It constructs bounded target-and-donor panels in memory from fixed seed streams. Its component, target, donor, and date identifiers have no physical-site identity or relation to AQS data. The execution entry point has no input-data path and is prohibited from reading AQS, v0.3.2, v0.4, taxonomy, API, or external files. This separation prevents an already viewed observational pattern from informing the v0.5 generator, calibration, threshold, policy, or interpretation.

The independent synthetic panels are not presented as a realistic physical model of PM2.5 transport or monitoring hardware. Their purpose is narrower: to create matched scope worlds that preserve the target path while controlling the comparison channel. The bounded raw-scale field and clipping-aware transformation are adverse tests of the analysis-scale algebra, not claims about the prevalence or magnitude of a real raw-scale process.

4.2 Primary public-data corpus

The primary corpus comprises the seven annual EPA AirData daily archives for parameter 88101, PM2.5 local conditions, spanning 2019–2025 [43, 45]. The data gate records the official download URL, archive bytes, archive SHA-256, selected CSV member, retained CSV row count, and acquisition-time modification field for every archive. This provenance record permits reconstruction without versioning the large raw archives or any API response in the repository.

The canonical signal is the reported `Arithmetic Mean` for `24-HR BLK AVG` observations. The predeclared cleaning rule keeps finite values with observation percent at least 75, a non-excluded Event Type, and a nonempty Method Code. It then refuses duplicate records with the same

Table 3: Frozen v0.3.2 data and audit inventory.

Quantity	Count
Canonical daily PM2.5 records	2,424,793
Monitor time series	1,689
Stable pseudo-anchors from Method Code regimes	124
Stable pseudo-anchors from all-monitor regimes	22
Persistent Method Code anchors	563
Anchors with at least one distinct physical donor	394
Anchors with at least three distinct physical donors	238
Complete common-method comparisons	228
Donor-insufficient anchors	325
Estimator input-window failures	10

Note. The physical donor identifier is State Code + County Code + Site Number; at most one POC is retained per donor site.

state, county, site, POC, and date key. Thus alternative daily summaries cannot be treated as independent observations. The retained primary inventory is a frozen snapshot; it is not a claim that the public source is immutable or complete for all later AQS updates.

The frozen deployment inventory contains 563 persistent anchors, of which 238 pass the distinct-physical-donor eligibility screen before the separate common-input-window check. These counts describe where the legacy observational workflow can be formed; they are not labels for local scope or physical mechanism.

The daily data are public, but raw archives and API responses are not included in this manuscript repository or its paper assets. The display-only case-study renderer verifies archive checksums locally before reading the same source files. That separation supplies a reproducible source chain without placing raw data, credentials, or local cache files under version control.

4.3 Metadata anchors and distinct physical donors

An anchor is a persistent reported Method Code transition with at least 60 calendar days on each side, at least 45 retained observations in each adjacent method run, and no more than a seven-day transition gap. Its fields preserve the old and new reported codes and method names, but neither field establishes why the record changed. An eligible geographic donor must be at a different physical site, within the predeclared 100-km radius, method-stable over the target window, observed on at least 60 paired pre-event days, and correlated at least 0.60 on the log scale before the anchor.

Physical donor identity is State Code + County Code + Site Number. A target's candidate donor list can contain multiple POCs for one physical site, but the ranked donor rule retains at most one: highest pre-event correlation, then shorter distance, then more paired days, then POC as a deterministic tie-breaker. This rule remediates an earlier superseded inventory in which multiple POCs could be treated as independent geographic donors. Same-site alternate POCs are excluded from geographic donor construction and reserved for contextual consistency analysis.

4.4 Stable-regime synthetic cases

Real anchors do not provide known physical-bias truth. The benchmark therefore constructs pseudo-anchors only inside stable target and donor method regimes, at least 60 days away from a reported target or donor transition. The frozen case manifest contains 124 method-transition stable regimes and 22 all-monitor stable regimes. Both sources are retained because the target

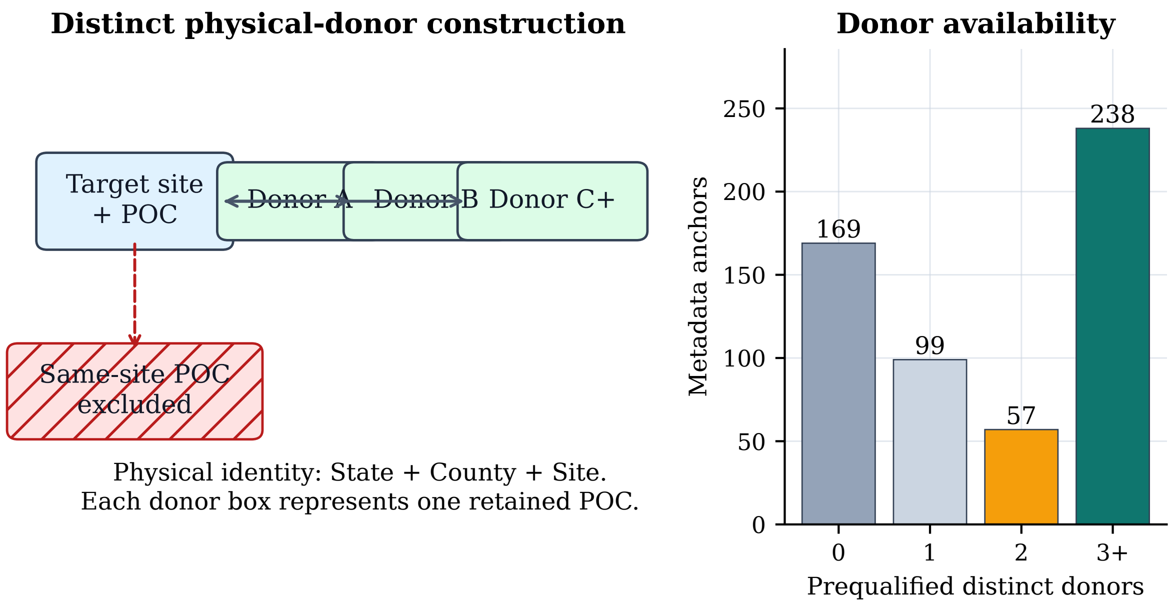

Figure 2: Distinct physical-donor construction and availability in the frozen 88101 inventory. A same-site alternate POC is explicitly excluded from the geographic donor set, and at most one ranked POC is retained at each other physical site. The right panel counts prequalified distinct donors before the common-method input-window check. These are audit availability counts, not counts of physical monitoring changes.

is a known, stable measurement record at the pseudo-anchor; neither is used as a real-event causal label. Section 6 explains how the target-plus-donor footprints are split before any held-out metric is calculated.

4.5 Audit disposition and parameter separation

Every anchor is retained in the event audit. A complete comparison means that the required common cross-site inputs exist; donor insufficiency and input-window failure are explicit machine-readable outcomes rather than omitted rows. This complete accounting is necessary because a result on only the well-matched subset would overstate the scope of cross-site inference.

Parameter 88502 is processed through a separate data gate, anchor inventory, and audit table. It is neither appended to nor pooled with the primary 88101 results. This separation prevents an apparent replication from being created by mixing parameter-specific reporting conventions, availability, or eligible event populations.

5 MetaShift-Bench audit framework

5.1 Channel-specific selective scope audit

The primary benchmark workflow is fixed before evaluation and does not read AQS data, v0.3.2 evidence, v0.4 outputs, or taxonomy decisions. It generates bounded synthetic target-and-donor panels in memory, assigns calibration and evaluation by independent component identifier and seed stream, then applies the complete Cartesian product of seven declared factors. Each local/shared pair reuses the same target path, target noise, component seed, base panel, and non-scope stress factors. The paired arms differ only in how much of the target schedule reaches available donors.

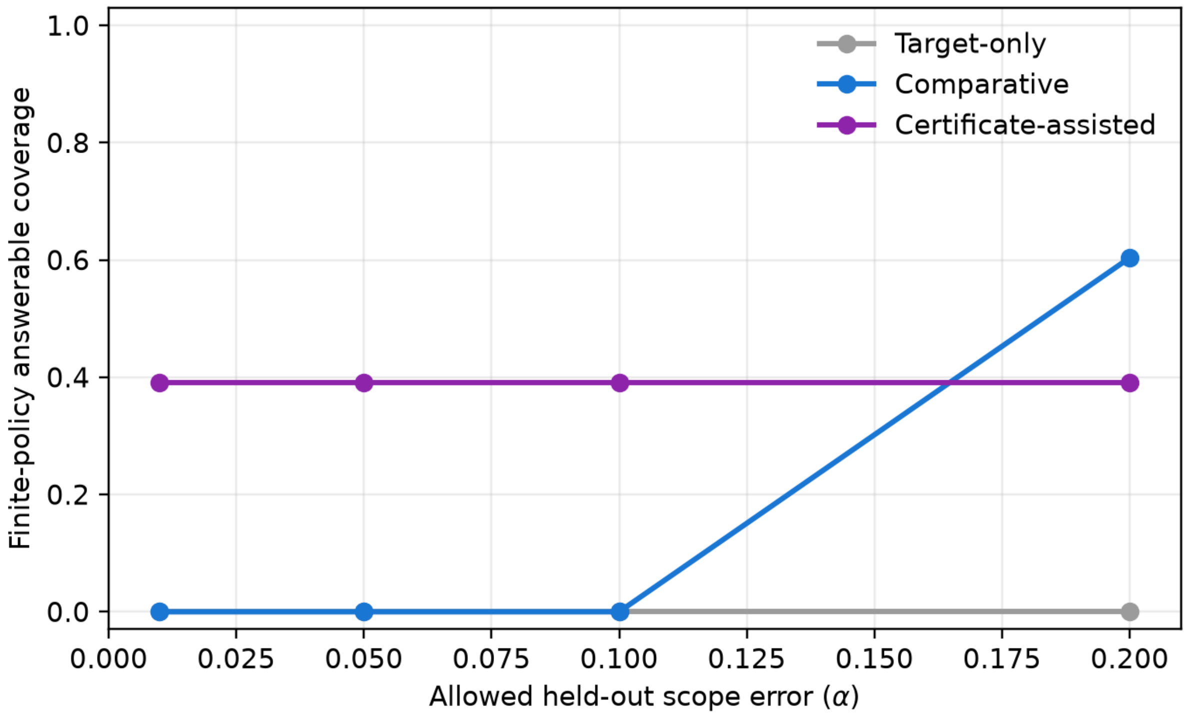

Figure 3: Receipt-verified v0.5 finite-policy Scope Answerability Frontier. Target-only and comparative policies have no positive-coverage qualifying rule through $\alpha = .10$. The comparative channel reaches coverage 0.603746 only at $\alpha = .20$, with observed conditional error 0.195208. The certificate-assisted policy uses synthetic design information and is displayed separately rather than treated as a deployable external-evidence channel.

The target-only forced policy receives a signed target-change score and always predicts local. The comparative forced policy receives the signed target-minus-availability-normalized-donor residual score and a threshold selected on calibration components only. Four confidence-selective policies use the same frozen prediction with separately calibration-chosen cutoffs for $\alpha = .01, .05, .10, .20$. The certificate-assisted policy is not a fitted confidence rule: it answers only when the declared structural margin is strictly positive. All thresholds, policy definitions, groupings, error tolerances, seeds, output schemas, and tests were fixed before the evaluation directory existed.

The complete held-out finite-policy envelope is reported in Table 5; its strict-tolerance failures remain part of the primary result rather than being hidden by the displayed comparative endpoint.

5.2 Partial-scope continuum and structural abstention

The five nominal q levels sample a partial-participation continuum rather than form a binary convenience sample. At $q = 0$, both donor and target observations are identical across local/shared arms and the stratum is a required comparative-channel negative control. At $q = 1$, every available donor receives the target schedule. For intermediate q, the fixed participation vector is renormalized over the actually available donor weights on each date. The design makes no claim that these five values are physical scope fractions in AQS data.

The structural margin aggregates a lower bound on the realized score gap and upper bounds for donor mismatch, bounded pre/post noise, nonlinear raw-scale leakage, and donor contamination. A nonpositive margin is an explicit structural abstention, not a confidence score rounded down after results are seen. The raw-field factor is intentionally outside the exact analysis-scale identity

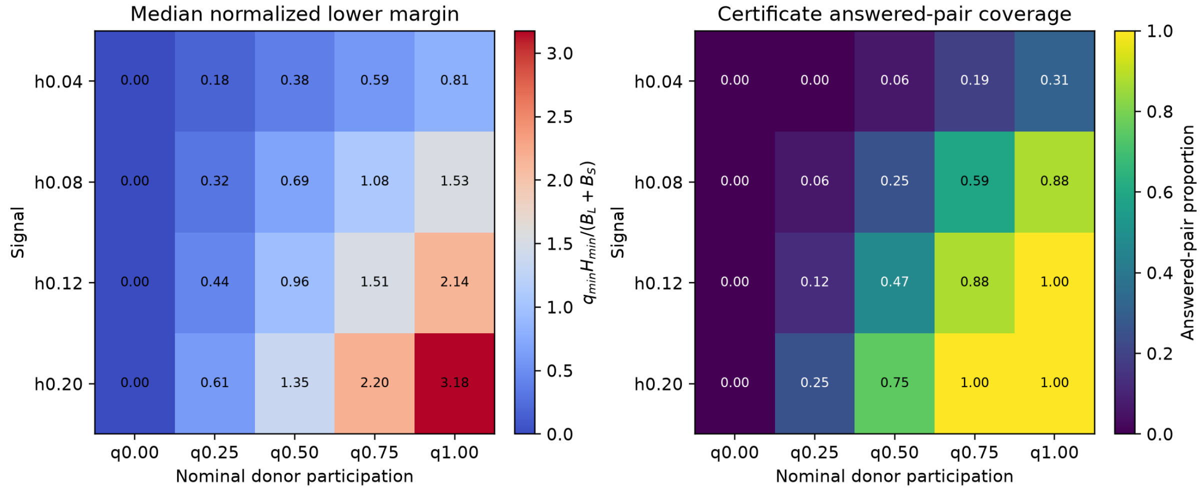

Figure 4: Normalized structural-margin phase diagram from the complete frozen v0.5 grid. The reference boundary at one separates declared interval separation from required certificate abstention. This is a property of the stipulated bounded generator and certificate inputs, not a map of real sensor network identifiability.

and is evaluated only through the clipping-aware bound.

5.3 Deployment audit framework

5.4 Counterfactual construction

For a complete anchor, the donor pool is ranked by pre-event correlation, distance, paired days, and POC only after physical-site deduplication. The nearest-neighbor difference-in-differences baseline places unit weight on the top-ranked donor. Standard synthetic control solves the nonnegative simplex problem

$$\min_{\substack{w_j \geq 0 \\ \sum_j w_j = 1}} \frac{1}{|T_{\mathrm{cal}}|} \sum_{t \in T_{\mathrm{cal}}} \left(z_{i,t} - \sum_{j \in N_a} w_j z_{j,t} \right)^2 .$$

The intercept is represented by calibration centering in the residual definition of Section 2. No post-anchor outcome is supplied to either method.

Fixed-weight MetaShift with a reliability-prior penalty starts from the pre-event reliability preference

$$\pi_{ij} \propto \max(\rho_{ij}, 0)^2 \exp(-d_{ij}/50),$$

where ρ_{ij} is the pre-event log-scale correlation and d_{ij} is distance in km. Its frozen real-audit objective is

$$\min_{\substack{w_j \geq 0 \\ \sum_j w_j = 1}} \mathrm{MSE}_{T_{\mathrm{cal}}}(z_i, Z_N w) + 0.1 \sum_j w_j^2 + 0.1 \sum_j (w_j - \pi_{ij})^2 .$$

This is a reproducible regularized weighting rule, not a learned instrument taxonomy. In the synthetic benchmark only, cross-validated MetaShift tests a frozen set of penalty pairs on the final 45 pre-event days, chooses the smallest validation RMSE, and refits the selected pair on the supplied pre-event calibration table. It never sees anchor-post data during selection.

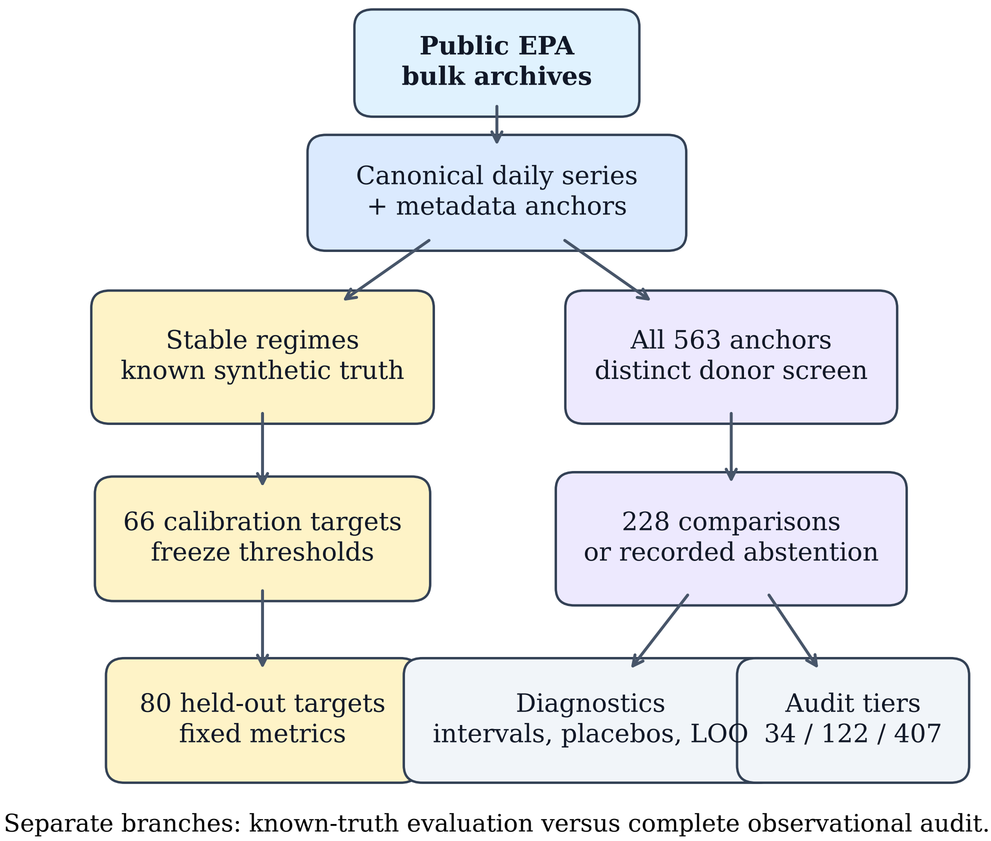

Figure 5: MetaShift-Bench evidence workflow. Public archives are canonicalized before persistent metadata anchors and distinct physical donor candidates are constructed. Stable regimes supply known synthetic truth, while every real anchor receives a common comparison or a recorded failure reason. The diagram is a workflow schematic; its arrows do not assert that metadata reveals a physical instrument state.

5.5 Two-level donor rule, algorithm, and computational scope

The audit uses two intentionally different donor conditions. An anchor is eligible for a common cross-site comparison only when at least three *distinct physical sites* pass the frozen geographic, stability, paired pre-event, and correlation screens. This protects against treating multiple POCs at a single location as independent geographic evidence. After dates are retained, a complete numerical comparison requires at least two available donors on a date and at least 30 retained observations in every required window. The latter condition protects calculation feasibility; it does not weaken the three-physical-donor eligibility rule.

For each anchor, the implementation follows the fixed sequence below:

1. identify and physically deduplicate prequalified donor candidates;
2. retain one deterministically ranked POC per donor site and abstain if fewer than three sites remain;
3. fit fixed cross-site weights using only the inclusive calibration slice, then center residuals on that same slice;
4. verify retained-date availability in calibration, pre, and post windows, recording an input-window failure when it is insufficient; and
5. compute the protocol-defined effect and diagnostic outputs only for a complete comparison.

If D candidates have T_{pre} paired pre-event records, screening and correlation assembly scale with the available paired data (on the order of DT_{pre}), and deterministic ranking adds $O(D \log D)$. The retained donor set is capped at three to five sites; residual assembly therefore scales linearly in the retained window length and this small set size. The constrained numerical fit has solver-dependent iteration cost, so the report does not claim a hardware-independent closed-form runtime. The complete audit ledger reports failures rather than treating an optimization or input failure as a negative physical finding.

5.6 Single-series comparators

The benchmark includes five transparent single-series baselines: before-after median, Bayesian mean shift, CUSUM, rolling-MAD, and PELT. Before-after median and Bayesian mean shift estimate a known-date 60-day level change. CUSUM, rolling-MAD, and PELT score or locate changes in the single target series [25, 32, 41]. They serve a necessary contrast: a method can detect an abrupt time-series pattern without establishing that the pattern is local rather than regional.

5.7 Interval and falsification diagnostics

For fixed donor weights, the primary interval resamples pre- and post-anchor residuals using circular moving blocks of seven days, with 1,000 fixed-seed repetitions. It is conditional on previously chosen donors and weights; it does not include selection uncertainty. The selection-aware nested procedure re-block-resamples the calibration matrix, recomputes correlation eligibility, selects three to five donors from the fixed candidate pool, refits the nonnegative MetaShift weights, and separately resamples the pre/post residual windows. It preserves temporal ordering and reports failed reselection or effect replicates rather than silently replacing them.

The audit also computes stable post-transition time placebos, donor-as-treated placebos, a date-resampling comparison, and leave-one-donor-out refits. These procedures are falsification and robustness diagnostics. They do not create random assignment, prove a physical mechanism, or make a residual effect a measured instrument bias.

Table 4: Predeclared primary evidence-tier rules and abstention outcomes.

Required diagnostic or condition	Threshold	Audit outcome
Complete common-method comparison	required	Missing input ⇒ inconclusive
Selection-aware nested interval	required	Unavailable ⇒ inconclusive
Unique stable post-transition time placebos	50	Fewer dates ⇒ inconclusive
Raw time-placebo probability	0.10	Above cutoff ⇒ not supported
BH-adjusted time-placebo q value	0.10	Unavailable ⇒ inconclusive; above cutoff ⇒ not supported
Fixed-weight conditional diagnostic interval	exclude 0	Otherwise ⇒ not supported
Leave-one-donor-out direction fraction	0.90	Below cutoff ⇒ not supported
All available required conditions	pass	Supported candidate discontinuity

Note. The rules synthesize observational diagnostics; they are not a supervised classifier, instrument-fault label, or physical-causality test. The nested interval is required for tier assignment but is not claimed to have synthetic coverage calibration.

5.8 Rule-based evidence synthesis and abstention

The tier logic distinguishes unavailable evidence from failed available diagnostics. An event is inconclusive if it lacks a common comparison, selection-aware interval, sufficient time placebos, an adjusted placebo quantity, or donor sensitivity. It is not supported when those required inputs are available but a pre-event fit, fixed diagnostic interval, placebo screen, or donor-direction condition fails. A supported candidate satisfies all available required conditions. Benjamini–Hochberg adjustment is used only as an exploratory multiple-screening quantity [12]; the resulting tiers are observational audit summaries, not labels for instrument failure, replacement, bias, or pollution correction.

6 Experimental design

6.1 v0.5 preregistered answerability-frontier experiment

The primary experiment is `v0.5-answerability-frontier`. It was implemented, tested, source-frozen, and independently protocol-verified before any evaluation output was created. It uses newly generated bounded synthetic panels only; AQS data and metadata, v0.3.2 evidence, v0.4 evidence, and the taxonomy review are forbidden inputs. Each component has 300 dates, one target, four donors, a fixed anchor day, fixed equal donor weights, and independently generated bounded common and idiosyncratic innovations. Calibration has 120 components, and evaluation has 360 components with 230,400 matched pairs and 460,800 scope arms; their identifiers and seed streams are disjoint.

For every component, the full $5 \times 4 \times 2^5 = 640$ Cartesian grid crosses nominal donor participation $q \in \{0, .25, .50, .75, 1\}$, signal strength $H \in \{.04, .08, .12, .20\}$, donor mismatch, availability loss, one-donor contamination, a common nonlinear raw-scale field, and bounded noise. Every cell produces one local and one shared arm. The target arrays are required to be exactly identical between the pair; at $q = 0$, donor arrays must also be identical. The protocol fails closed on target-identity violations, footprint leakage, stale hashes, incomplete cells, calibration/evaluation mixing, or outcome-dependent configuration changes.

The frozen design yields 76,800 calibration pairs and 230,400 evaluation pairs, with 460,800 evaluation scope arms and 614,400 total scope arms. A calibration-only threshold selects the comparative forced policy. Confidence cutoffs are selected from 101 calibration quantiles separately for each predeclared calibration tolerance. Evaluation then reports every frozen policy at every reporting tolerance; it does not select a different policy or retune a cutoff after outcomes. The certificate threshold is analytic and uses frozen synthetic-design inputs: the

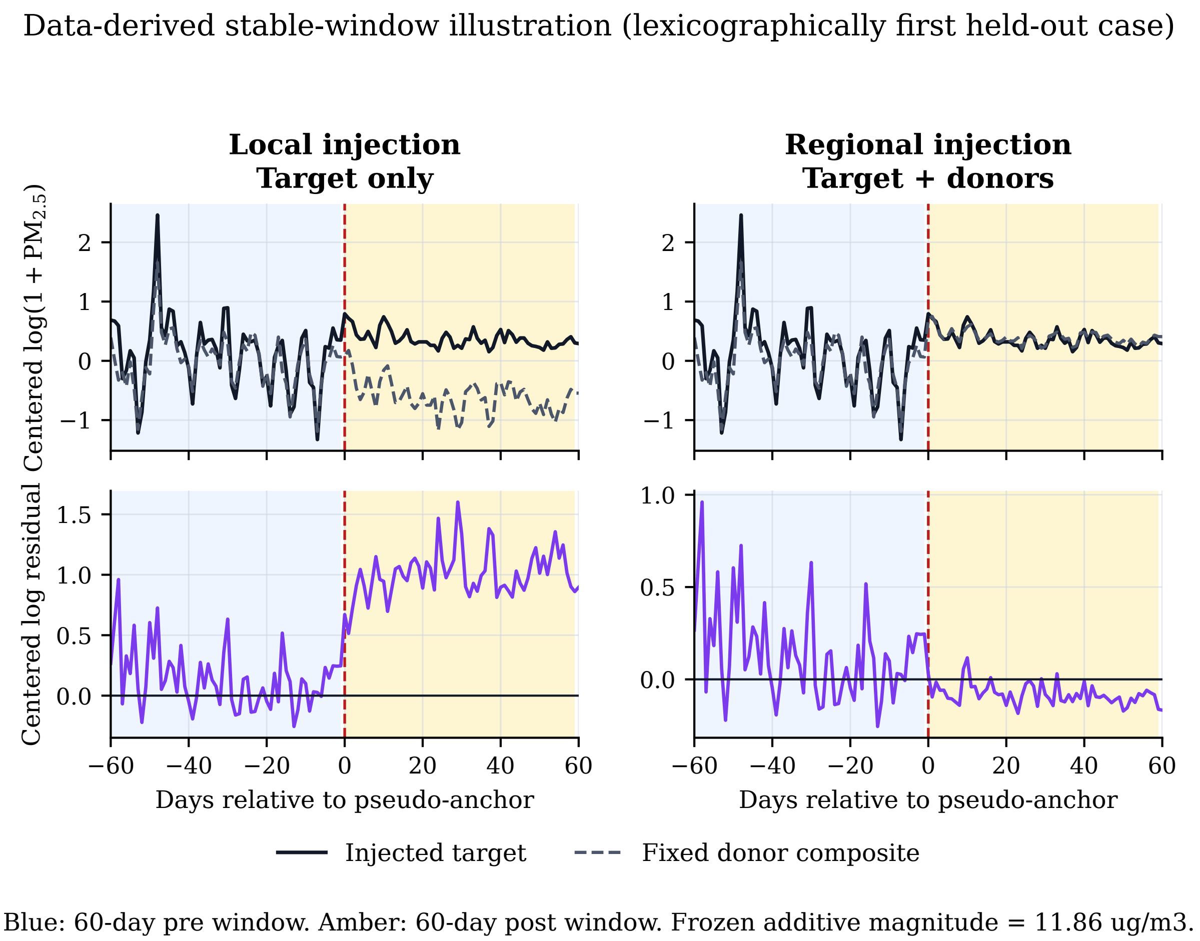

Figure 6: Data-derived stable-window illustration for the lexicographically first held-out v0.3.2 case fixed in the display contract. The same frozen additive injection is applied to the target alone (left) or to the target and donor composite (right). The illustration is historical endpoint context, not the v0.5 partial-scope continuum or a model of a real Method Code transition.

declared bounds plus the realized retained-date availability and panel values required by the clipping-aware raw-field bound. The ambiguous partial region consists of all $q > 0$ pairs with nonpositive structural margin, where the certificate must abstain; $q = 0$ is recorded separately as observationally unanswerable.

The unit of uncertainty is the evaluation component, not a scope arm, date, or grid row. The protocol uses a 1,000-repetition percentile component bootstrap with prespecified seed and reports full policy risk–coverage, envelope, certificate, and subgroup results without collapsing them into a favorable score. The source freeze tag, remote execution-claim tag, manifest hashes, and receipt are described in Section 11. The single authorized execution is not retuned or rerun after its results are inspected.

6.2 Historical v0.3.2 stable-regime synthetic benchmark

Real metadata anchors have no physical-bias ground truth. Known-effect evaluation therefore uses only stable target and donor regimes separated from nearby reported transitions. The frozen `stable_full_v2` manifest contains 146 physical target sites. Whole connected components of the complete target-plus-donor physical input graph were assigned to the 66-site calibration subset or the 80-site held-out evaluation subset, so no physical input site crosses the split.

The benchmark injects target-only additive, proportional, gradual-drift, temporary-step, and

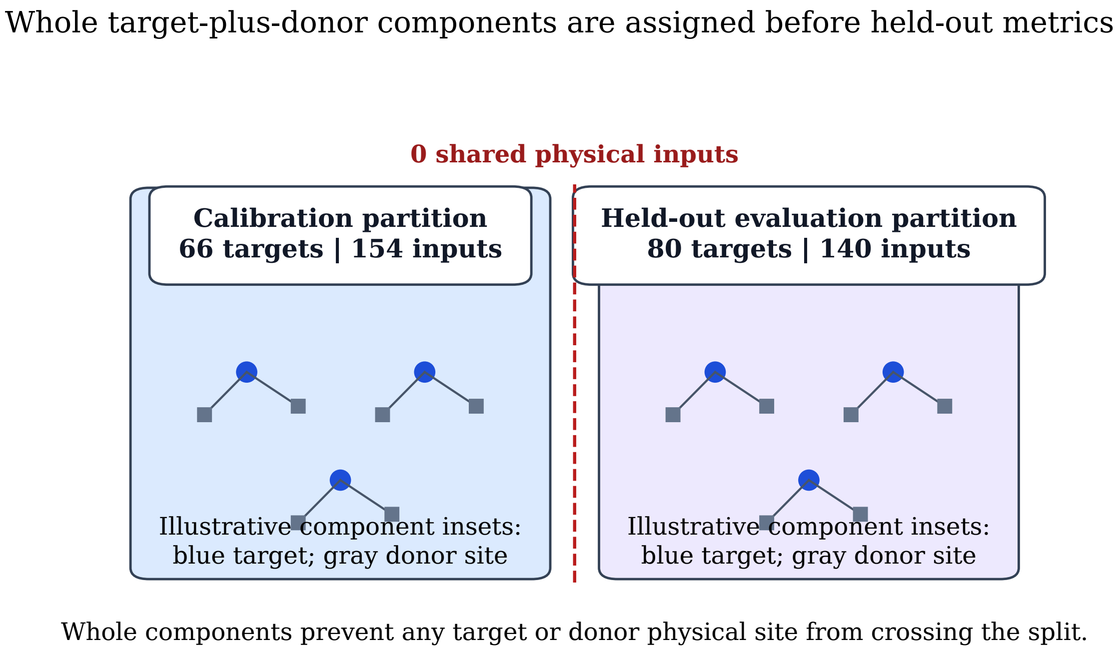

Figure 7: Leakage-resistant stable-case split, including the complete target-plus-donor physical input footprint rather than target sites alone. Whole connected components are assigned to calibration or held-out evaluation before metrics are computed; the frozen split audit records zero shared physical inputs. The small networks are explanatory insets, not a geographic map. This controls physical-input reuse across partitions; it does not make the synthetic perturbations a model of every real monitoring process.

variance changes, together with matched target-and-donor regional variants. Each perturbation variant contains 400 held-out evaluation samples. Each family therefore combines a local condition and a matched regional condition; the family rows in the results are not local-only effect estimates. Five frozen magnitude multipliers (0.5, 1.0, 1.5, 2.0, and 3.0) scale an additive step, proportional step, gradual drift, temporary step, or variance increase. The variance-only condition has no defined level-effect truth, so local-effect MAE is correctly reported as N/A for that condition.

The effect shape, magnitudes, seeds, candidate rule, model settings, and metrics were frozen before evaluation. For every method, a ranking threshold is selected only by macro-F1 on the calibration partition and then applied unchanged to the evaluation partition. The viewed evaluation partition is not used for model, threshold, donor-weight, quality-gate, or taxonomy adjustment. This is a protocol safeguard, not a claim that one finite split eliminates all future generalization uncertainty.

6.3 Benchmark metrics and uncertainty

For target-only non-variance perturbations, local-effect MAE is the mean absolute difference between the estimated and injected 60-day median log effect. AUPRC measures ranking of local against matched regional variants; macro-F1, precision, recall, and regional false-positive rate use the calibration-only threshold. CUSUM, PELT, and rolling-MAD have no protocol-defined effect-magnitude estimate, so their MAE cells are N/A while their ranking metrics remain observable.

The MetaShift-versus-standard-MAE comparison uses a paired cluster bootstrap over stable case IDs with 1,000 repetitions and fixed seed 20260830. Pairing keeps a method comparison tied to the same pseudo-anchor, perturbation, magnitude, and injected seed. It reports uncertainty for the frozen benchmark comparison, rather than a universal probability that one method dominates another.

6.4 Real-anchor audit and sensitivity

The real-anchor workflow uses the same distinct-physical-donor construction, pre-event fitting boundaries, and effect window across the three common cross-site methods. The five single-series baselines use the same 60-day before/after window but do not receive a geographic counterfactual. The audit records complete comparisons and explicit non-complete dispositions for all 563 anchors. Cross-validated MetaShift is evaluated only in the synthetic benchmark; it is not introduced into the real-audit comparison after observing those events.

Predeclared sensitivity analyses vary one factor at a time: donor-screening rules, 45/60/90-day effect windows, raw versus log reporting scale, and evidence-tier thresholds. The sensitivity rows reveal availability and interpretation dependence; they are not a search for the setting with the most favorable residual pattern. The standard synthetic-control rows in the main and reliability-ablation experiments are checked for exact alignment to tolerance 10^{-10}, preventing a seed or input mismatch from masquerading as an ablation finding.

6.5 Interval coverage protocol

Fixed-weight interval coverage is evaluated on held-out synthetic effects. The confidence level, seven-day block length, perturbation types, and 1,000 repetitions are fixed in the frozen configuration. The separate full selection-aware synthetic coverage protocol was frozen before outcomes. Its feasibility assessment records that full donor re-screening and weight refitting at the specified nested scale is infeasible within the submission deadline. The report therefore does not convert real-event selection-aware intervals into coverage-calibrated physical-bias intervals.

6.6 External context and reproducibility

Same-site alternate-POC records, narrow hourly POC windows, QA availability, and targeted official-document review supply contextual checks with known limitations. A separate parameter-88502 pipeline uses its own scan and audit; it is never pooled with 88101. The frozen release records the exact source commit, locked Python environment, source manifests, machine-readable checks, and a two-environment comparison of designated core artifact hashes.

7 Results

7.1 RQ0: Scope Answerability Frontier and information gain

Answer. The frozen v0.5 finite-policy results support the stipulated target-only impossibility endpoint and show that comparative answerability is not uniformly available at strict error tolerances. The target-only forced policy answers every held-out arm (coverage 1.000000) and has observed conditional scope error 0.500000; accordingly, no positive-coverage target-only policy qualifies for any reported $\alpha < .5$. The comparative forced policy also answers every arm (coverage 1.000000) but has observed error 0.275095. Its confidence-selective policies calibrated at .01, .05, and .10 answer 0 evaluation arms. Only the policy calibrated at .20 reaches a positive comparative envelope: coverage 0.603746 at observed error 0.195208.

Table 5: Frozen finite-policy scope-answerability envelope on the held-out v0.5 evaluation components. Parentheses give observed conditional scope error for the policy attaining the displayed coverage; – means no positive-coverage candidate qualified. The certificate-assisted channel uses synthetic design information and is not an operational deployment channel.

Tolerance α	Target-only	Comparative	Certificate-assisted
0.01	0.000000	0.000000 (–)	0.390612 (0.000000)
0.05	0.000000	0.000000 (–)	0.390612 (0.000000)
0.10	0.000000	0.000000 (–)	0.390612 (0.000000)
0.20	0.000000	0.603746 (0.195208)	0.390612 (0.000000)

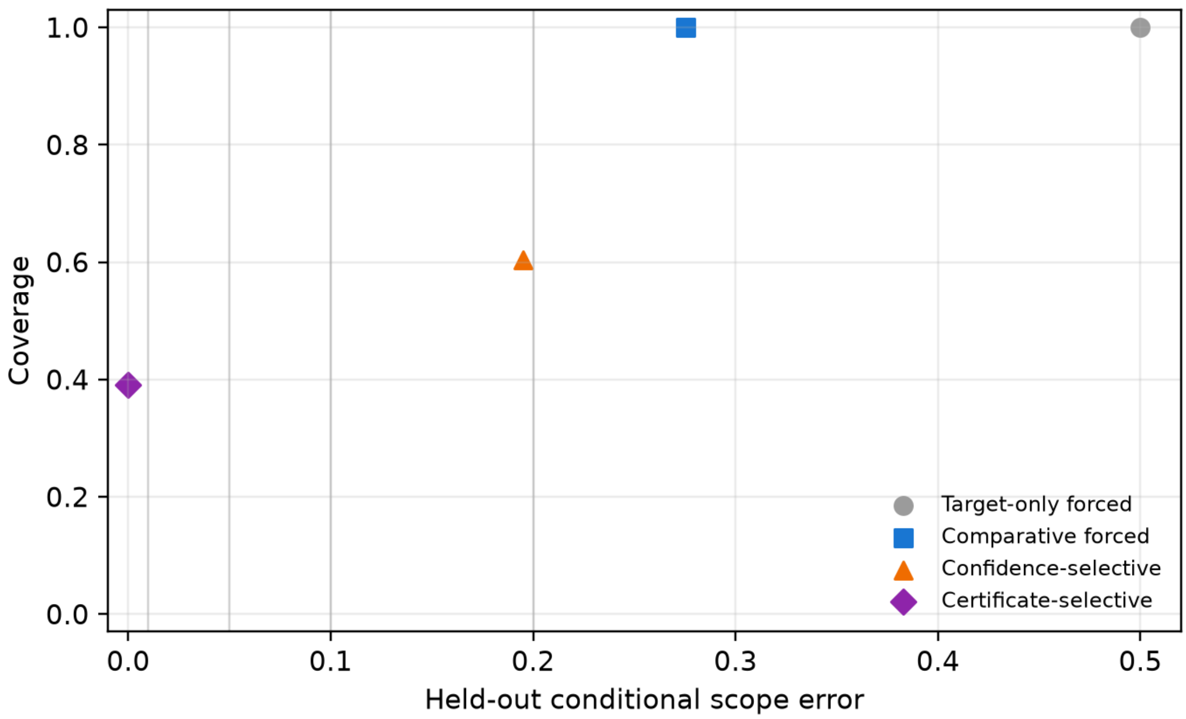

Figure 8: Receipt-verified held-out v0.5 risk–coverage comparison for all predeclared policies. Forced comparative classification has full coverage but substantial error. Strict confidence policies abstain entirely; the least strict frozen confidence policy answers at its calibration target but remains above the stricter reporting tolerances. The certificate is shown separately because it uses synthetic design information.

At $\alpha = .20$, the achieved finite-policy Scope Answerability Gain from the target-only to comparative channel is therefore 0.603746. Its 95% component-bootstrap interval is $[0.600863, 0.607008]$. At .01, .05, and .10, both finite-policy frontiers are zero. These values compare frozen policies on held-out synthetic components; they do not estimate a universal optimum or guarantee future conditional scope risk.

7.2 RQ0a: Structural certificate and retained failure boundary

Answer. The bounded certificate is internally valid within its declared synthetic contract, but it is intentionally incomplete. It answers 89,997 of 230,400 matched pairs (0.390612) and has 0.000000 observed conditional error among 179,994 answered scope arms. Its predeclared simulation-information-oracle efficiency is 0.976583. It abstains on 140,403 pairs with nonpositive structural margin and has 0 observed envelope violations. These are empirical observations under a bounded generator, not evidence of zero population risk or a deployable real-network certificate.

Table 6: Certificate behavior by nominal donor participation on held-out v0.5 pairs. Efficiency is coverage relative to the predeclared simulation-information oracle region; it is undefined at $q = 0$ because that negative-control stratum has no oracle-answerable pairs.

Participation	Pair coverage	Conditional error	Oracle efficiency
$q = 0.00$	0.000000	–	–
$q = 0.25$	0.109375	0.000000	0.875000
$q = 0.50$	0.382812	0.000000	0.942308
$q = 0.75$	0.663997	0.000000	0.988435
$q = 1.00$	0.796875	0.000000	1.000000

Certificate validity from frozen evaluation

Group	Certificate coverage	Conditional error	Envelope violations	Efficiency vs oracle
overall	0.391	0.000	0.000	0.977
q0.00	0.000	-	0.000	-
q0.25	0.109	0.000	0.000	0.875
q0.50	0.383	0.000	0.000	0.942
q0.75	0.664	0.000	0.000	0.988
q1.00	0.797	0.000	0.000	1.000

Figure 9: Certificate validity summary from receipt-verified v0.5 outputs. The certificate has no observed envelope violations under the stipulated bounded construction, but its coverage and oracle efficiency remain condition-dependent. It is a synthetic-design-assisted sufficient condition, not an empirical external-evidence policy.

The $q = 0$ stratum contains 46,080 matched pairs with exactly identical target and donor observations in both scope arms. It produces 0 certificate answers and no oracle-answerable pair, as required by the negative-control contract. For $q > 0$, the certificate coverage rises with nominal participation but decreases under contamination, nonlinear raw fields, and heteroskedastic noise. The result does not imply that all real partial scope is objectively ambiguous; it shows where this particular structural bound refuses to answer.

7.3 Historical v0.4 endpoint sanity check

The immutable v0.4 construction supplies only an endpoint check: when matched local and regional worlds hold the target sequence fixed, target-only classification has observed error 0.5,

Table 7: Predeclared failure and abstention accounting on 230,400 held-out v0.5 matched pairs. These rows are retained rather than converted into a favorable single score.

Outcome	Count	Interpretation
Target and donor observations identical at $q = 0$	46,080	negative-control abstention stratum
Nonpositive structural margin	140,403	certificate must abstain
Certificate-answered pairs	89,997	0 errors among 179,994 answered arms
Envelope violations	0	none observed
Comparative-forced errors	126,764	among 460,800 forced scope arms
Confidence policy at calibration $\alpha = 0.05$	0	no answered arms

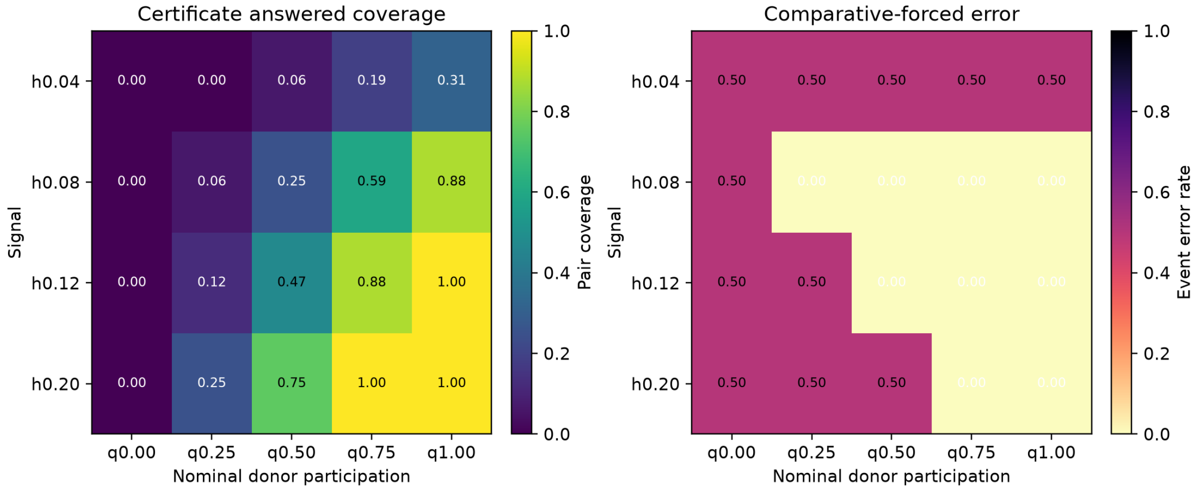

Figure 10: Complete v0.5 failure-mode map. Low signal and adverse comparison conditions retain forced-comparative errors or certificate abstentions instead of being excluded. The $q = 0$ row is an observational-identity negative control, not comparative success.

Table 8: Independent stable-regime synthetic evaluation.

Method	MAE	AUPRC	Macro-F1	Regional FPR
Standard SC	0.09983	0.91476	0.81641	0.140
MetaShift fixed	0.10344	0.89832	0.79515	0.150
MetaShift CV	0.09742	0.90178	0.80916	0.135
Nearest-neighbor DiD	0.11150	0.88443	0.78852	0.075

Note. Metrics are aggregates over the frozen `stable_full_v2` evaluation set. Lower MAE and regional false-positive rate are preferable; higher AUPRC and macro-F1 are preferable.

while its constructed comparative rule has zero observed error on that finite endpoint evaluation. It does not measure a risk–coverage frontier, calibration-selected abstention, partial scope, a failure boundary, or real measurement mechanisms. The v0.5 continuum is therefore the primary scope evidence; v0.4 is retained as an historical sanity check and is not rerun or reinterpreted.

7.4 RQ1: Stable-window synthetic comparison

Historical answer. The frozen v0.3.2 benchmark distinguishes cross-site methods from single-series baselines, but no method is uniformly best across all metrics and paired perturbation families. Table 8 reports the four cross-site methods used for the primary comparison, while Table 9 restores all nine frozen methods. Figure 11 visualizes the main cross-site trade-offs without collapsing them into a post-hoc composite score.

Standard synthetic control has stronger aggregate attribution ranking and macro-F1 than both MetaShift variants, while cross-validated MetaShift has the smallest aggregate local-effect MAE. Nearest-neighbor difference-in-differences has the lowest regional false-positive rate among these four methods but weaker effect error and attribution metrics. These are benchmark-specific trade-offs, not a ranking for every monitoring network.

The single-series methods remain in the complete comparison because an anchor date alone can make a time-series break appear compelling. Their inclusion shows what is lost when a method does not ask whether target and donors moved together. N/A local-effect MAE cells are retained rather than replaced by a surrogate metric: CUSUM, PELT, and rolling-MAD supply change evidence but no protocol-defined level-effect estimator for the variance-only condition.

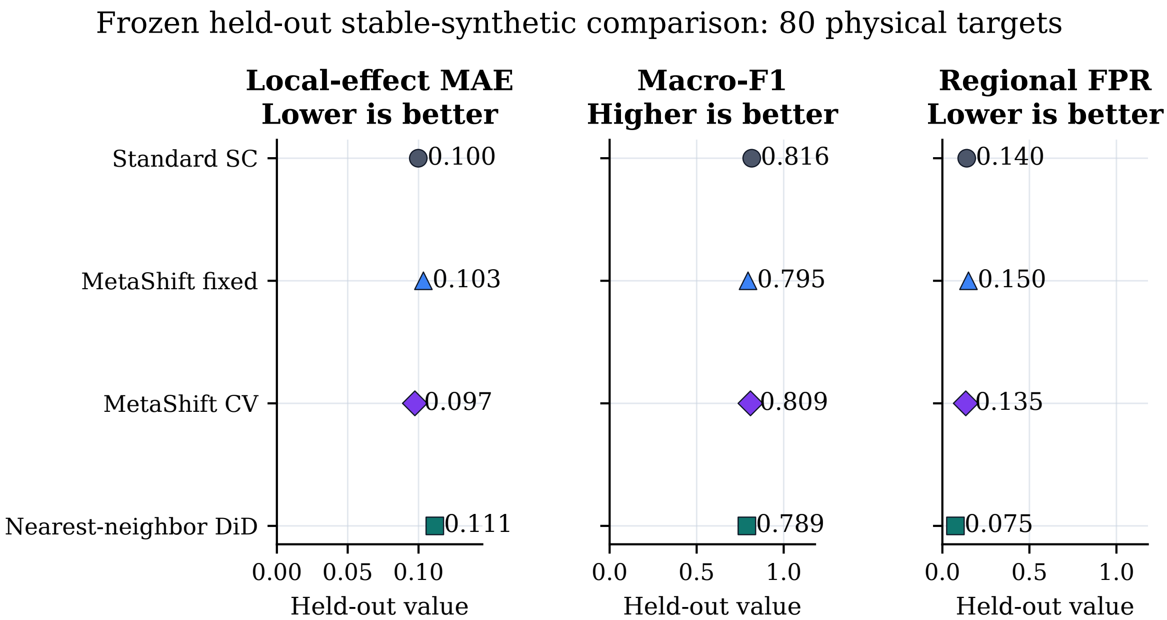

Figure 11: Held-out stable-regime synthetic comparison for the four cross-site methods. Dots show local-effect MAE, macro-F1, and regional false-positive rate on the frozen 80-site complete-input-disjoint partition; AUPRC is retained in Table 8. Lower MAE and regional false-positive rate are preferable, whereas higher macro-F1 is preferable. The panels do not form a post-hoc composite ranking; all values originate in the saved `stable_full_v2` metrics table.

Table 9: Complete frozen aggregate comparison of all benchmark methods.

Method	Local-effect MAE	AUPRC	Macro-F1	Regional FPR
Standard SC	0.09983	0.91476	0.81641	0.140
MetaShift fixed	0.10344	0.89832	0.79515	0.150
MetaShift CV	0.09742	0.90178	0.80916	0.135
Nearest-neighbor DiD	0.11150	0.88443	0.78852	0.075
Bayesian mean shift	0.24672	0.50116	0.49962	0.542
Before-after median	0.25330	0.50125	0.50024	0.505
CUSUM	N/A	0.50117	0.49720	0.581
Rolling MAD	N/A	0.50112	0.49427	0.402
PELT	N/A	0.50025	0.49976	0.522

Note. All values are aggregates over the held-out stable-regime evaluation partition. Lower MAE and regional FPR are preferable; higher AUPRC and macro-F1 are preferable. N/A means that the method supplies no local-effect magnitude estimate for that perturbation design, not that the method was omitted.

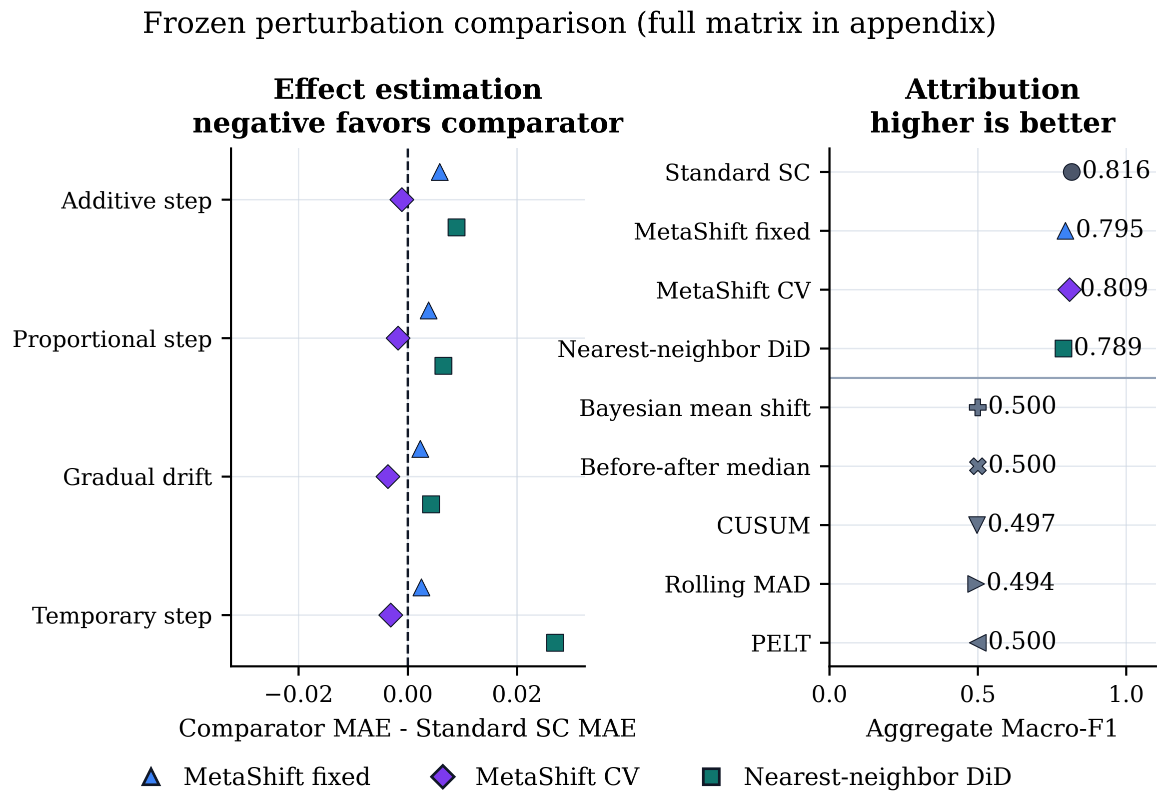

Figure 12: Main-text perturbation comparison. The left panel gives each non-variance perturbation family's local-effect MAE difference from standard synthetic control for the three cross-site comparators; negative values favor the comparator on that metric. The right panel retains aggregate macro-F1 for all nine frozen methods, separated by cross-site versus single-series status. The complete five-family, all-metric matrix remains in Table 21; variance-only local-effect MAE is N/A by definition, not an exclusion.

Table 10: Paired event-cluster bootstrap for local-effect MAE differences.

Comparison	Difference	95% bootstrap interval	Clusters
MetaShift fixed minus Standard SC	0.00361	[-0.00401, 0.01163]	80
MetaShift CV minus Standard SC	-0.00240	[-0.00772, 0.00347]	80

Note. Difference is MetaShift minus standard synthetic control; negative values favor MetaShift on MAE. Both bootstrap intervals include zero.

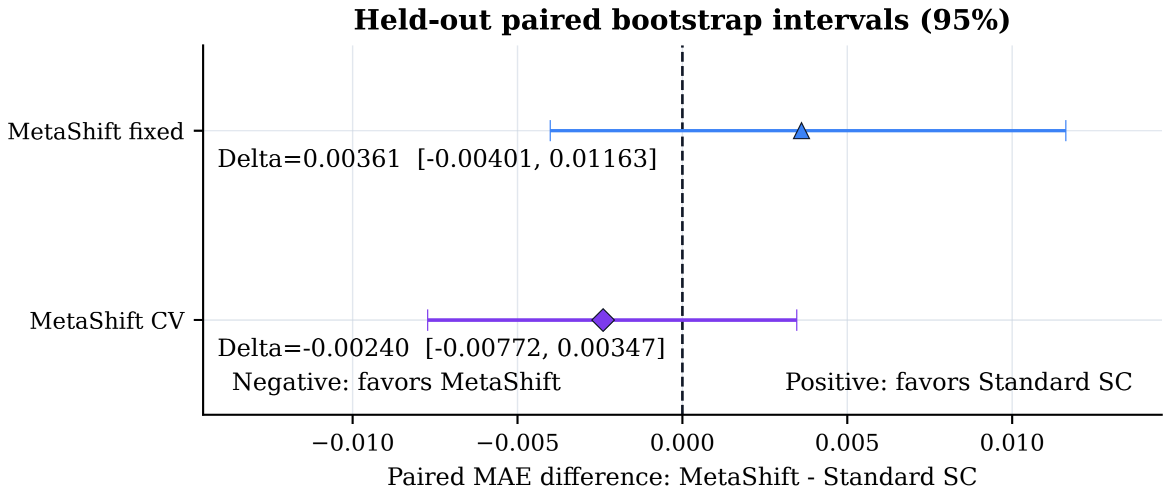

Figure 13: Paired event-cluster bootstrap intervals for MetaShift minus standard synthetic-control local-effect MAE on matched held-out synthetic instances. The dashed vertical line is no difference; negative values favor MetaShift on MAE. Both frozen intervals cross zero, so the figure does not support a confidence-supported aggregate superiority claim.

The perturbation matrix makes the scope of the aggregate result visible. Cross-site methods distinguish additive and gradual target-local changes much better than the single-series baselines under this design, whereas proportional, temporary, and variance conditions expose different false-positive and ranking trade-offs. Because each row aggregates a paired local and regional condition, it tests local-versus-regional discrimination rather than a generic detection-rate claim.

7.5 RQ2: Bound on a MetaShift superiority claim

Answer. Neither MetaShift variant has a confidence-supported aggregate MAE advantage over standard synthetic control. The paired bootstrap intervals cross zero for both frozen variants, including cross-validated MetaShift's -0.00240 difference with interval [-0.00772, 0.00347].

The main and ablation experiments share 7300 standard synthetic-control records exactly to the configured numerical tolerance. Table 11 therefore compares reliability-prior variants without changing the standard-control reference rows. The ablations do not establish a stable overall MetaShift advantage; this negative result closes the model-superiority route for the frozen benchmark.

7.6 RQ3: Complete real-anchor audit

Answer. Only 228 of 563 anchors support a complete common cross-site comparison under the distinct-physical-donor rule. The remaining 325 donor-insufficient and 10 input-window events are retained as applicability boundaries rather than removed.

Table 11: Reliability-prior and regularization ablations on shared synthetic inputs.

Variant	MAE	Macro-F1	Regional FPR
Standard SC	0.09983	0.81641	0.140
No ridge penalty	0.10289	0.80127	0.138
Ridge = 0.01	0.10304	0.80263	0.130
Add coverage	0.10332	0.79862	0.135
MetaShift full prior	0.10344	0.79515	0.150
No reliability-prior penalty	0.10350	0.80167	0.133
No correlation term	0.10362	0.79858	0.133
No distance term	0.10370	0.80588	0.088
Uniform prior	0.10378	0.80521	0.112
Ridge = 1.0	0.10500	0.80285	0.097
Direct reliability	0.10543	0.80574	0.105

Note. The common standard-synthetic-control records align exactly across the main and ablation experiments to tolerance 10^{-10}.

Table 12: Complete 88101 event audit and real-event diagnostics.

Audit disposition	Anchors
Complete common-method comparison	228
Fewer than three distinct physical donors	325
Estimator input-window failure	10
Total	563

Method	Events	Fixed conditional interval excludes 0	Median log effect
MetaShift fixed	228	159	-0.08704
Standard SC	228	145	-0.07078
Nearest-neighbor DiD	228	134	-0.07490

Note. Nested selection-aware intervals are available for 227/228 complete events; 153 exclude zero. Their mean width is 0.19057 log units versus 0.17863 for fixed-weight MetaShift intervals. Leave-one-donor-out refits complete for 227 events, with 202 retaining direction under every donor removal. All intervals are diagnostic, not calibrated physical-bias confidence intervals.

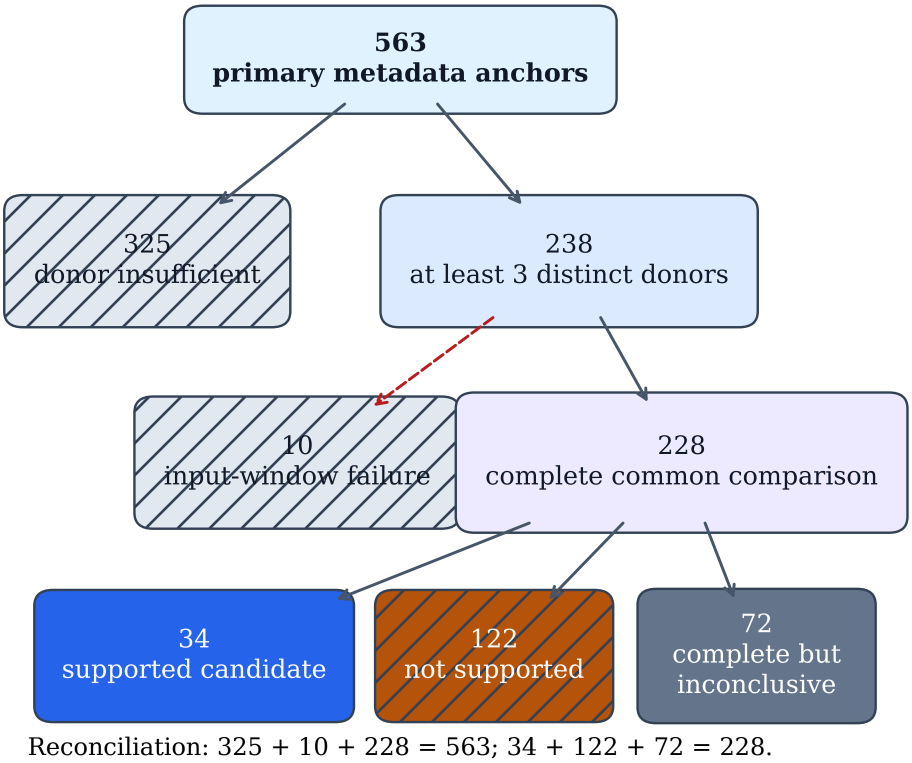

Figure 14: Hierarchical accounting of all 563 primary metadata anchors. Donor-insufficient and input-window-failure rows remain inconclusive audit dispositions. Only complete comparisons flow into the observational three-way tier rules; the lower reconciliation confirms both count partitions. No branch represents a physical monitoring diagnosis.

Table 13: Placebo, external-document, and same-site POC evidence boundaries.

Diagnostic	Value
Complete time-placebo events with at least 50 dates	157
Complete time-placebo events with 100 dates	128
Raw time-placebo probabilities at most 0.10	71
BH q values at most 0.10	40
Donor-as-treated placebo records	866
Date-resampling permutations	200
Dated site-specific external confirmations	0/20
Usable hourly same-site POC comparisons	9/11
Daily/hourly POC direction agreements	8/9
Standard SC 90th-percentile risk-gate retention	75/80

Note. Time and donor placebos are diagnostic falsifications. Same-site POC and document review provide contextual evidence only, not instrument ground truth.

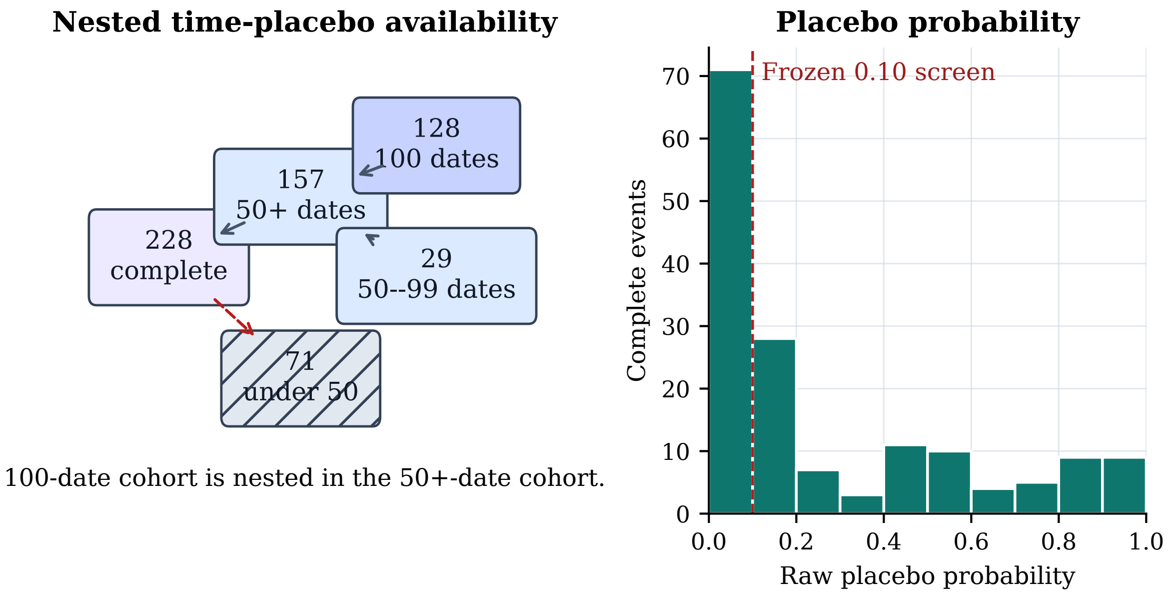

Figure 15: Time-placebo availability and raw within-event placebo-probability distribution. The vertical line marks the frozen exploratory 0.10 screening threshold. These values are diagnostic falsification quantities, not causal probabilities; incomplete placebo sets remain visible in the left panel.

The exploratory evidence synthesis assigns 34 supported candidates, 122 not-supported events, and 407 inconclusive events. The supported label means that predeclared diagnostic conditions were jointly met; it does not confirm a physical failure, replacement, or causal bias.

Time-placebo coverage is available for 157 complete events with at least 50 dates, including 128 with 100 dates. The donor-as-treated table contains 866 rows, and the 200-resample date comparison has upper-tail probability 0.00498. These are diagnostic falsifications rather than causal probabilities.

7.7 RQ4: Interval coverage and sensitivity

Answer. The two available fixed-protocol interval constructions do not support a generic calibration claim. At nominal 95%, fixed-weight conditional block-bootstrap coverage ranges from 63.875% to 67.281% over 3,200 held-out effect instances per method. At nominal 90%, split-conformal coverage ranges from 98.8125% to 99.5625% and is highly conservative.

Selection-aware nested intervals exist for 227 complete real events and exclude zero for 153.

Table 14: Held-out synthetic interval coverage by method.

Method	Fixed 95% coverage	Fixed mean width	Conformal 90% coverage	Conformal mean width
Standard SC	67.281%	0.265	98.8125%	0.396
MetaShift fixed	63.875%	0.270	99.4375%	0.471
MetaShift CV	65.406%	0.265	98.9375%	0.437
Nearest-neighbor DiD	66.656%	0.287	99.5625%	0.565

Note. Each method has 3,200 evaluation effect instances. Fixed-weight intervals under-cover and split-conformal intervals over-cover under this frozen synthetic design.

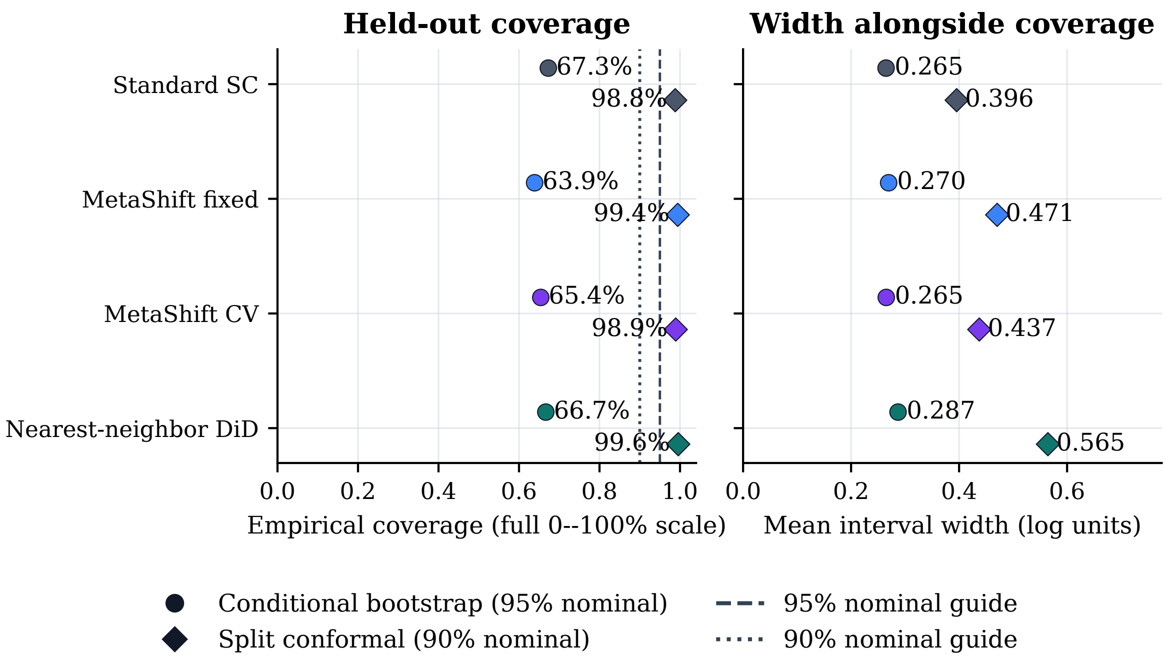

Figure 16: Empirical held-out synthetic coverage and mean interval width for four cross-site methods. Circles are fixed-weight conditional moving-block intervals at nominal 95%; diamonds are split-conformal intervals at nominal 90%. The coverage panel intentionally uses the full 0–100% scale, and paired width values show the cost of conservative coverage. Dashed reference lines identify nominal targets. Undercoverage and overcoverage are negative calibration findings, not parameters tuned after inspection.

Table 15: Predeclared effect-window and reporting-scale sensitivity.

MetaShift window	Complete events	Median log effect	Sign agreement with 60 days
45 days	224	-0.08769	93.3%
60 days	228	-0.08704	reference
90 days	196	-0.08064	93.9%

Method	Log/raw sign agreement	Spearman ρ
MetaShift fixed	95.2%	0.829
Standard SC	92.5%	0.868
Nearest-neighbor DiD	92.1%	0.843

Note. Window sensitivity adds a full method-stability check for every target and donor window. Reporting-scale agreement is a concordance diagnostic and does not equate raw and log causal estimands.

Table 16: One-factor screening sensitivity with three required distinct donors.

Setting	Eligible before donor rule	With at least 3 donors
Primary (100 km)	563	238
Coverage 70%	563	243
Coverage 80%	563	232
Stable window 45 days	572	252
Stable window 90 days	543	214
Transition gap 3 days	512	212
Transition gap 14 days	589	254
Donor radius 50 km	563	102
Donor radius 200 km	563	426
Correlation $\rho \geq 0.50$	563	241
Correlation $\rho \geq 0.70$	563	230

Note. One predeclared setting changes at a time. These counts describe audit availability, not detected-effect frequency.

Each requested 1,000 repetitions and retained at least 521 valid repetitions. This result does not validate their coverage: the frozen full selection-aware synthetic coverage protocol is infeasible within the deadline. The intervals are therefore reported as conditional diagnostics.

Table 15 reports predeclared effect-window and reporting-scale sensitivity; Table 16 reports one-factor screening availability. The geographic donor radius visibly changes how many anchors can be analyzed, so these settings are documented rather than optimized after viewing results.

7.8 RQ5: External context and transfer boundary

Answer. External evidence supplies limited contextual support, not ground truth. Of 11 same-site POC candidates, 9 have usable paired hourly windows and 8 agree in daily/hourly direction. The targeted official-document review found 0/20 dated site-specific confirmations. A missing public notice is unavailable evidence, not evidence that no physical change occurred.

The separately pinned QA-collocation review contains 12 candidate records, of which 2 match the target POC and 0 provide adequate matched pre/post windows. It is therefore an evidence-availability result, not a validation of a measurement-bias hypothesis.

The independent 88502 pipeline has 34 eligible metadata anchors and only 3 complete common-method comparisons. It demonstrates separate-pipeline feasibility but is too sparse to support broad generalization. EPA documentation about network data alignment provides useful method context [42]; it is not a dated site-specific confirmation for the reviewed anchors.

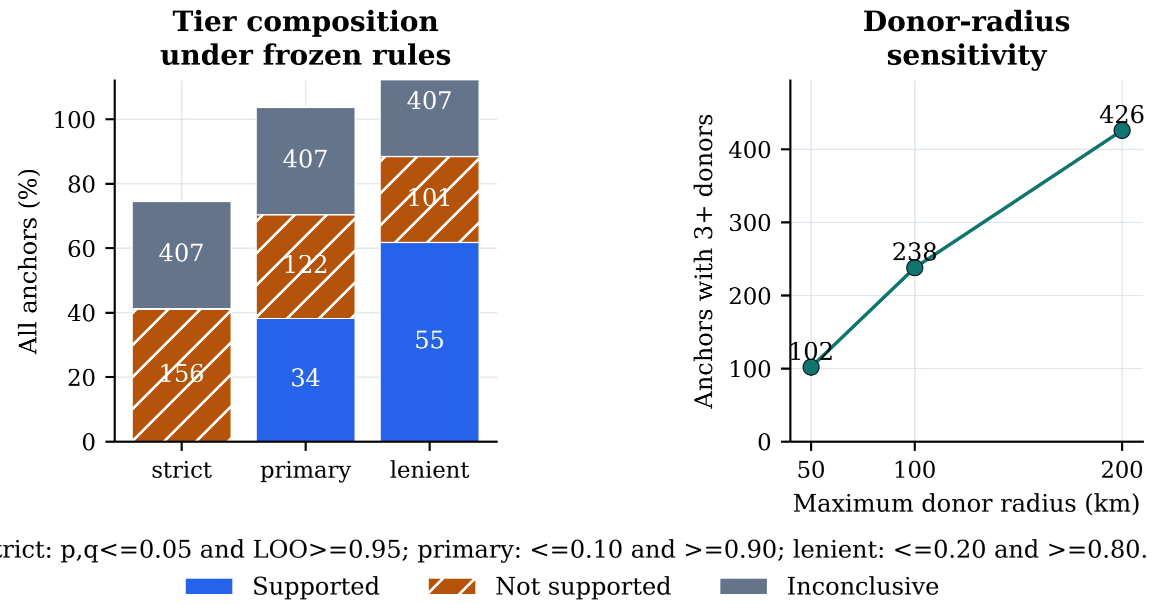

Figure 17: Predeclared evidence-tier and geographic-screening sensitivity. The left panel normalizes tier composition to all anchors while retaining counts in the bars; the right panel shows how the three-donor availability boundary changes with the predeclared radius. These describe rule dependence and data availability, not physical causality or a search for a favorable specification.

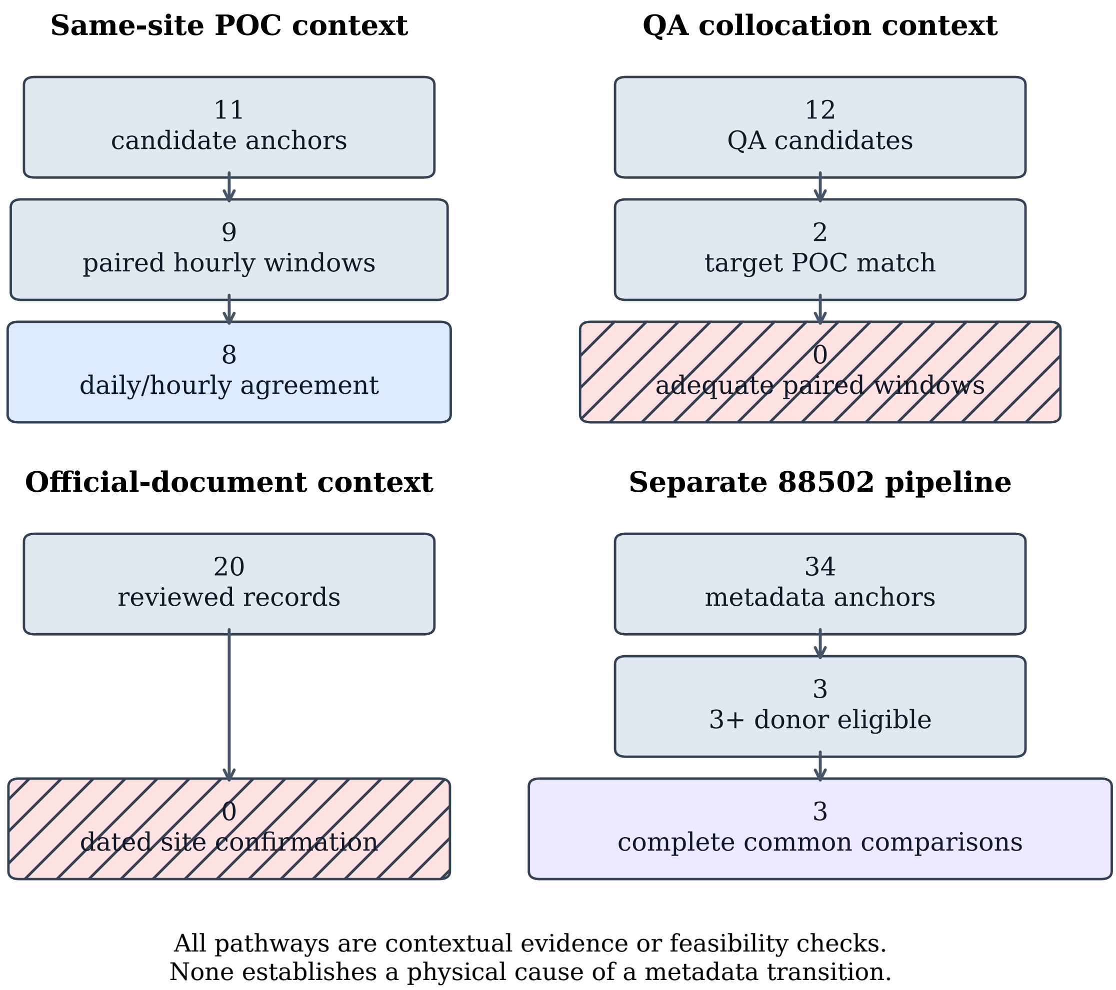

Figure 18: Four contextual-evidence ladders: same-site POC, QA collocation, targeted official-document review, and the separate 88502 pipeline. Each ladder visibly retains unavailable evidence, including zero adequate QA matched pre/post windows and zero dated site-specific confirmations. Neither a POC agreement nor an unlocated document supplies universal physical-instrument ground truth.

8 Representative case studies

8.1 Selection policy and source provenance

Examples can make an audit readable, but selected examples can also conceal a failure mode or overstate a method's scope. The three cases in this section are therefore not hand-picked for their visual contrast or effect magnitude. The display configuration selects one complete supported-candidate anchor and one complete not-supported anchor by minimum distance to the within-tier median standardized score, with anchor-ID tie-breaking. It selects the lexicographically first donor-insufficient inconclusive anchor to make the absence of a qualified counterfactual visible. The configuration, selected donors, archive manifest checksum, donor-table checksum, fitted weights, and reconstruction error are stored in the generated case-study manifest.

For the two complete cases, the renderer reads only the checksum-pinned public EPA archives, applies the frozen cleaning rule, reconstructs fixed-weight MetaShift with the reliability-prior penalty from pre-anchor data, and fails if the reconstructed saved log effect differs by more than 10^{-9}. The raw archives remain outside version control. The third case intentionally has no reconstructed counterfactual: its audit disposition is that fewer than three qualified distinct physical donors were available.

8.2 Interpretation boundaries

The supported-candidate panel illustrates a case in which the predeclared diagnostic conditions jointly hold. That status prioritizes the anchor for further station-history and technical-record review; it does not identify a device, a maintenance action, a calibration change, or a measurement bias. The not-supported panel illustrates the opposite distinction: a visible target series and a complete counterfactual are not sufficient when available diagnostics fail the frozen evidence rules. It would be misleading to discard that case merely because its residual plot is less dramatic.

The inconclusive panel is equally important. A target series may have an apparent before/after pattern, yet the protocol cannot form the common cross-site comparison if the distinct-donor requirement fails. MetaShift-Bench therefore abstains rather than borrowing a same-site POC, relaxing the donor rule after inspection, or assigning a physical explanation. This case makes the audit's coverage boundary concrete: lack of a qualified comparison is lack of evidence for this protocol, not evidence that nothing happened.

Table 17: Deterministically selected representative audit cases.

Case	Metadata anchor	Disposition	Saved diagnostic summary
Supported candidate	`12-057-0113 poc1-2023-12-01`	Complete	Effect -0.12777; fixed [-0.17258, -0.10511]; nested [-0.17399, -0.10560]
Not supported	`06-031-0004 poc8-2021-01-12`	Complete	Effect -0.08240; fixed [-0.21609, 0.03012]; nested [-0.24309, 0.03662]
Inconclusive	`01-103-0011 poc3-2023-02-01`	Donor insufficient	No counterfactual, effect, or interval is imputed.

Note. Supported and not-supported cases minimize absolute distance to their within-tier median standardized score, with anchor ID tie-breaking. The inconclusive case is the lexicographically first donor-insufficient anchor and deliberately has no counterfactual or interval. Intervals are diagnostic, not calibrated physical-bias confidence intervals.

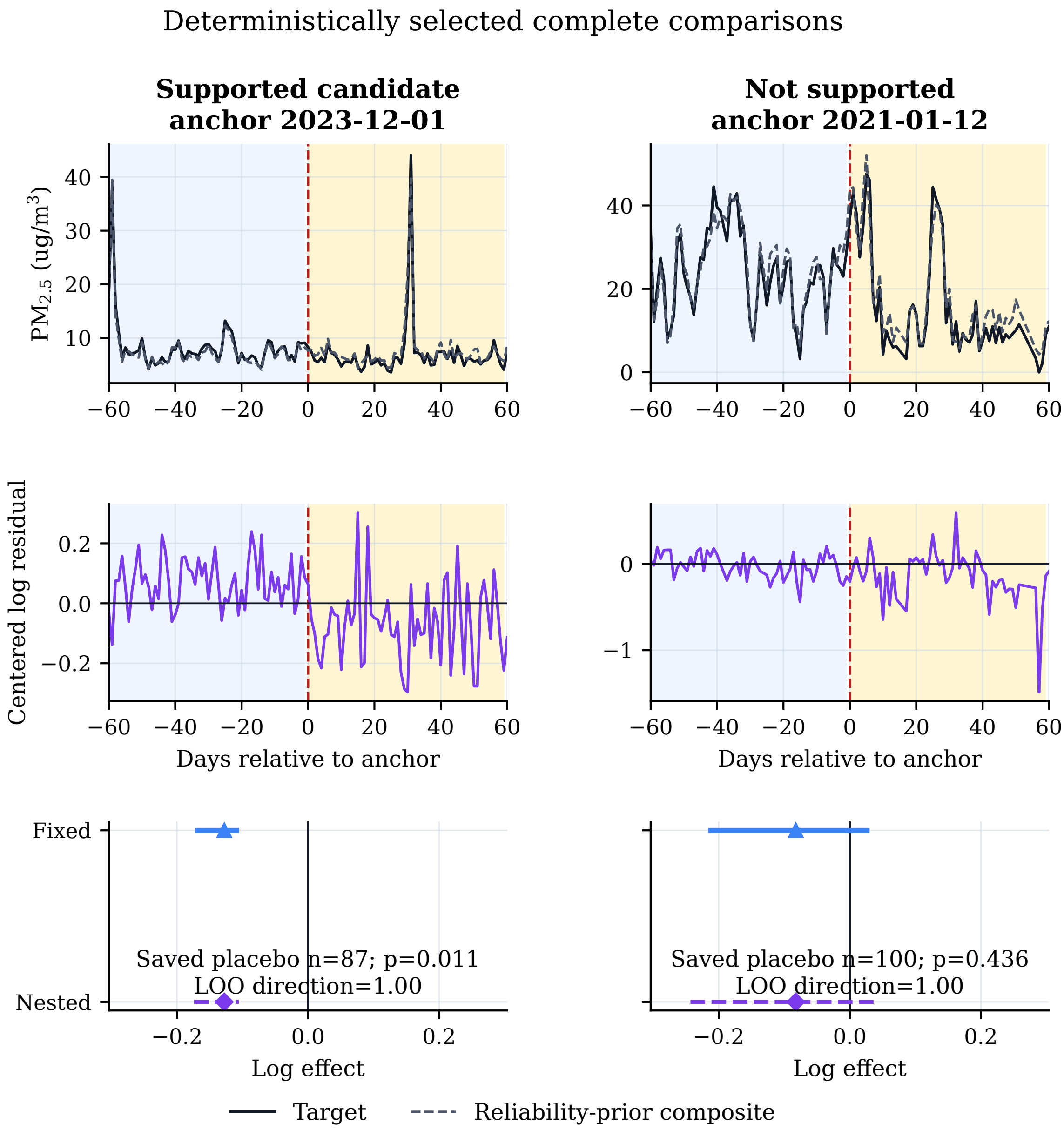

Figure 19a: Deterministically selected complete representative cases. The top row shows retained daily target values and the pre-anchor-fitted reliability-prior MetaShift counterfactual in μg/m^3; the middle row shows calibrated log residuals; and the lower row shows saved placebo summaries with fixed and nested diagnostic intervals. Blue and amber shading mark the 60-day pre- and post-anchor comparison windows. The figure displays saved diagnostic outputs, not evidence of a physical instrument event.

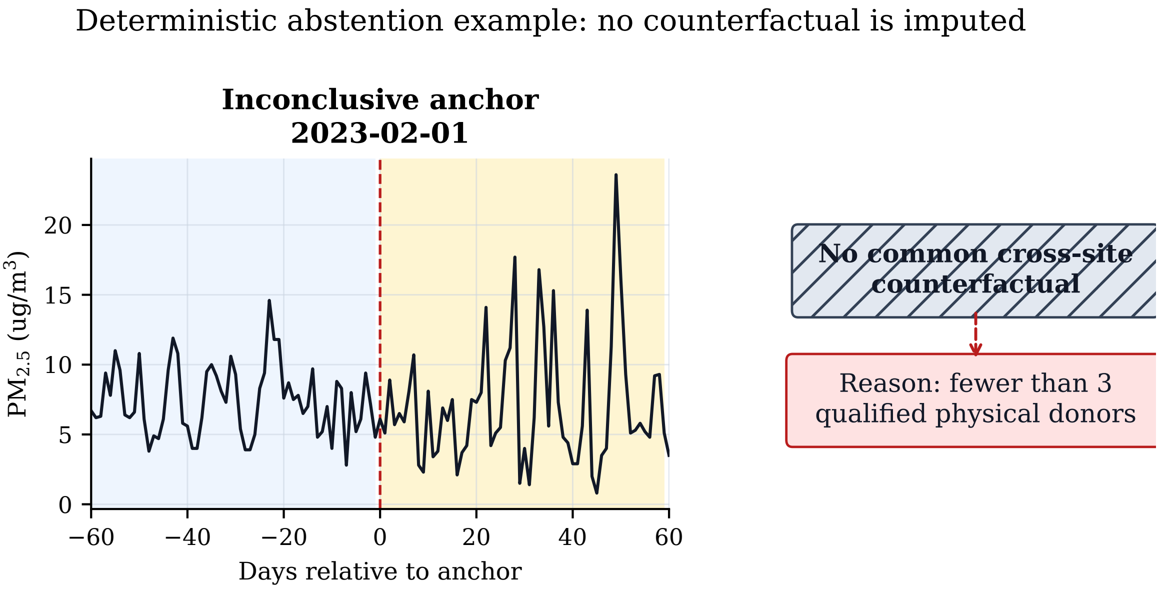

Figure 19b: Deterministic representative audit abstention. The retained target series is shown without a constructed counterfactual because fewer than three qualified distinct physical donors were available. The protocol therefore records an unavailable common comparison and does not impute a placebo, effect, or interval.

9 Discussion

MetaShift-Bench reframes the central computer-science question as *answerability*, not estimator superiority. The v0.5 result makes clear why this distinction matters. The target-only channel is structurally uninformative for balanced matched scope labels, and the comparative channel does not magically remove that limitation: its frozen finite-policy frontier is zero at the three strict tolerances. At $\alpha = .20$ it has positive coverage, but that policy's observed answered-case error remains close to the tolerance. This is useful evidence precisely because it rules out an unqualified claim that a donor residual always resolves scope.

The structural certificate provides a different kind of result. Given synthetic design information and bounded envelopes, it safely refuses nonseparated cases and had no observed envelope violations in this execution. Its coverage is intentionally lower than the comparative confidence policy, and its efficiency is measured only against a predeclared simulation-information oracle. Neither the certificate nor the oracle is a physical mechanism model, a confidence calibration theorem, or a deployable monitoring-network policy. The right conclusion is a conditional boundary: when the stated structural assumptions hold, the score intervals separate; when they do not, abstention is mandatory.

This hierarchy clarifies the roles of the project's earlier evidence. The v0.4 local/regional construction established a useful exact endpoint but could not characterize how information degrades between endpoints. The v0.3.2 AQS release retains value as a reproducible deployment and abstention audit: it shows how often a distinct-physical-donor comparison is computationally available and how external records remain insufficient for mechanism claims. Neither predecessor validates v0.5 synthetic scope labels as real-world scope truth.

The broader evidence architecture still matters. A compelling graph can conceal a non-comparable donor, an unrecorded selection decision, or an unavailable input window. A useful benchmark should make those contingencies visible rather than treating them as implementation details.

The physical-site/POC distinction is a concrete example. An AQS POC identifies a reported occurrence in the data system, not a universally stable physical instrument identity. If multiple POCs from one physical site are counted as independent donors, a target can appear to have multiple controls while its geographic evidence is not genuinely independent. The v0.3.2 donor rule therefore chooses at most one POC per physical donor site by a deterministic pre-event ranking. This choice reduces nominal availability, but it makes the meaning of a "three-donor" comparison honest. The many donor-insufficient anchors are not a defect to hide: they quantify where a cross-site method is not supported by the available data under the chosen independence proxy.

The stable-regime synthetic experiment supplies the one part of the project with known injected truth. Its target-only and matched regional perturbations directly test the computational distinction that a raw change-point does not: is a pattern specific to the target relative to comparable donors, or shared with them? The full method table makes clear why a single headline metric would be inadequate. Cross-site procedures can exploit reference information, while single-series procedures remain useful guards against overclaiming because they show what can be detected without a regional comparison. The perturbation-family matrix also avoids presenting a favorable aggregate as evidence that one method works equally well for additive, proportional, temporary, drifting, and variance-only patterns.

The result for MetaShift is deliberately negative. Cross-validated MetaShift has a favorable point estimate for local-effect MAE, but the paired event-cluster bootstrap interval against standard synthetic control crosses zero. Standard synthetic control also leads the frozen aggregate in attribution ranking and macro-F1. It would be scientifically inappropriate to convert one point estimate, selected experiment, or visually appealing case into a broad algorithm-superiority statement. The report preserves this result in the abstract, results, conclusion, source validator, and claim ledger. It also avoids further optimization on the already viewed held-out partition. A future model proposal would require a new physical-footprint-disjoint blind manifest, frozen parameters, and a precommitted metric set before outcomes are viewed.

The real-anchor audit has a different role from the synthetic benchmark. It does not validate a model against physical measurement-bias truth, which is unavailable from the Method Code field. Instead, it answers a bounded operational question: for which reported transitions can the frozen counterfactual and diagnostic workflow be executed, and what does each available diagnostic say? Complete comparisons, donor insufficiency, and input-window failures are all part of the result. This complete accounting guards against survivorship bias from reporting only well-matched stations. It also separates a necessary computational abstention from a substantive claim about the monitoring site.

The evidence tiers implement the same separation at a finer level. A supported candidate has passed a conjunction of predeclared observational diagnostics, including pre-fit quality, a conditional fixed interval, time-placebo screens, and leave-one-donor-out stability. It is not a supervised label and it does not prove a physical fault. A not-supported event has the required evidence but fails at least one diagnostic; an inconclusive event lacks a required comparative or diagnostic input. This three-way output is more informative than an all-or-nothing detector because it shows whether a low-confidence result comes from contradictory diagnostics or from unavailable data. It is also a safer decision interface for follow-up work: only humans with station records can decide whether a candidate merits an equipment-level hypothesis.

The interval results are another reason to avoid overly strong language. Fixed-weight moving-block intervals under-cover the known synthetic effects at their nominal level, while split-conformal intervals over-cover in the same frozen design. Neither observation can be corrected by post hoc adjustment without invalidating the held-out evaluation. The nested real-event procedure does capture some uncertainty from time resampling, donor re-selection, and weight refitting inside the observed candidate pool, but full selection-aware synthetic coverage at that scale was frozen as infeasible within the deadline. Thus real-event intervals are useful protocol-defined diagnostics, not calibrated confidence intervals for measurement bias. This distinction is important because numerical intervals can otherwise imply more physical certainty than their construction supports.

External context adds perspective but not ground truth. Same-site POC and hourly comparisons may reveal directional consistency, yet POCs are not universal equipment identities and temporal agreement alone cannot establish mechanism. The targeted public-document review likewise records an evidence shortfall rather than interpreting an unlocated notice as proof that no change occurred. The separate 88502 pipeline demonstrates that the machinery can be applied without pooling parameter codes, but its sparse complete-comparison population limits any transfer conclusion. These results support a workflow in which automated evidence synthesis guides a records search rather than replaces it.

Finally, reproducibility is part of the scientific result. The frozen release binds source artifact hashes, input-footprint checks, document checks, claim-to-evidence mappings, and independent environment hashes. The presentation layer adds deterministic figures and checksum-pinned case reconstruction without modifying the analysis artifacts. These safeguards cannot establish biological, engineering, or causal truth. They do make it possible for another investigator to distinguish a reproducible computation from an unsupported interpretation, which is the appropriate achievement for a metadata-anchored audit benchmark.

10 Limitations and threats to validity

The benchmark is intentionally designed to expose limitations rather than to hide them behind a binary classifier. Table 19 organizes the residual risks by construct, internal, statistical, and external validity. The mitigations constrain interpretation, but none turns the audit into a physical instrument diagnosis.

10.1 Scope-answerability assumptions

Table 18 separates the assumptions used by the v0.5 statements from claims that they do not support. A violation of the certificate contract is not a low-confidence answer: it invalidates the certificate for that pair and requires abstention. In particular, bounded innovations and known synthetic error terms are design choices; they cannot be silently replaced by unbounded real-world noise models.

Table 18: Assumptions and boundaries for v0.5 scope-answerability claims.

Assumption	Supports	Does not support
Balanced labels, matched target laws, and no label-dependent target side channel	Target-only impossibility in the stipulated synthetic experiment.	Target-only impossibility for arbitrary priors, leaked features, or real AQS records.
Nested target-only and comparative channels under the same law	Policy-set monotonicity of the population frontier.	Strict empirical improvement by a particular finite-sample classifier.
Analysis-scale interventions, fixed weights, and arm-invariant retained dates	Exact $\overline{q_t h_t}$ score separation.	Exact raw-scale algebra after nonlinear transformation or clipping.
Declared bounded mismatch, noise, leakage, and contamination envelopes	Sufficient interval separation when $\kappa > 0$.	A confidence guarantee, unbounded-noise result, or real sensor certificate.
Synthetic design information for participation and bounds	Certificate coverage and oracle-efficiency diagnostics.	An empirical O_E external evidence channel or a deployable policy.

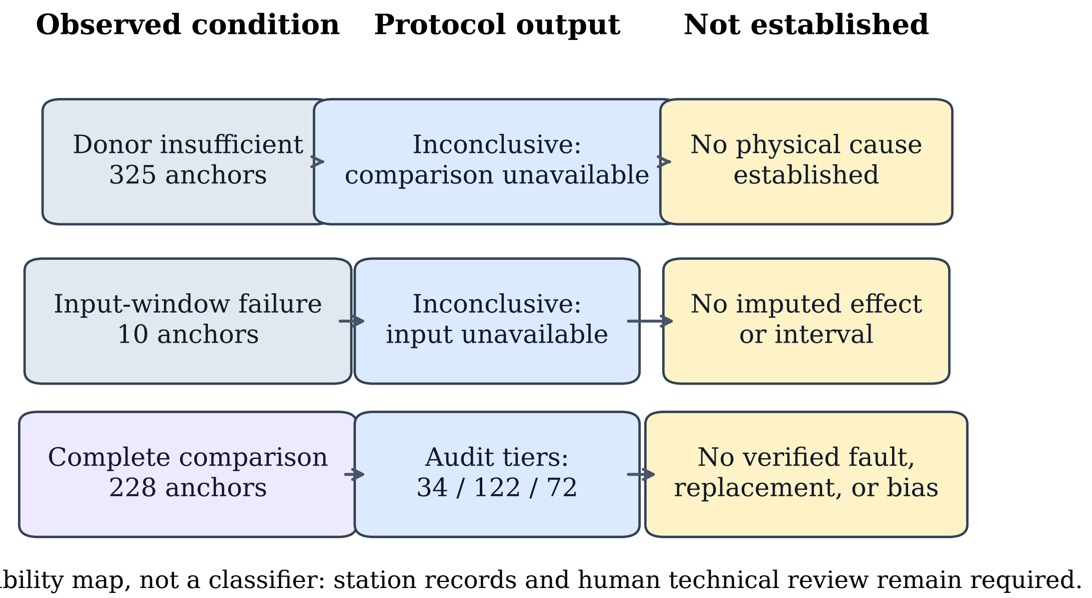

Figure 20: Applicability and failure-mode map for the frozen audit. The diagram is not a classifier: an available cross-site diagnostic result and an inconclusive availability failure both leave physical mechanism unresolved. Station histories, technical records, and human review remain necessary for any equipment-level hypothesis.

Table 19: Threats to validity, mitigation, residual consequence, and needed future evidence.

Validity class	Risk and current mitigation	Residual consequence	Needed future evidence
Construct	**Risk.** A Method Code or POC may not encode a physical device, calibration, siting, or maintenance event. **Mitigation.** Treat records as metadata anchors; prohibit fault, replacement, bias, and correction claims.	Residual effects cannot identify a physical mechanism.	Dated station histories, calibration records, deployment logs, and independently reviewed taxonomy.
Synthetic scope	**Risk.** The v0.5 local/shared labels, bounded panel model, and synthetic design-information certificate may not represent operational networks. **Mitigation.** Target-fixed pairing, complete-grid accounting, separate negative controls, and explicit certificate nonclaims.	Finite-policy frontiers and zero observed certificate error do not establish population or real-network scope accuracy.	Independent blind generators, naturally occurring labeled scope events, and admissible external records.
Internal	**Risk.** Weather, smoke, emissions, processing, or donor instability can create a target-reference residual. **Mitigation.** Pre-event matching, distinct physical donors, time and donor placebos, and leave-one-donor-out checks.	Diagnostics can reject some explanations but do not establish randomized causal attribution.	Meteorological covariates, emissions records, network operations logs, and prospective comparator studies.
Statistical	**Risk.** Donor selection, serial dependence, multiplicity, and incomplete resampling can invalidate naive uncertainty. **Mitigation.** Fixed conditional and nested intervals, moving blocks, exploratory BH screening, and reported coverage failures.	Real-event intervals are diagnostic rather than calibrated estimates of physical bias: they are not an estimate of a causal physical bias or coverage-calibrated physical-bias confidence intervals.	A computationally feasible, predeclared full selection-aware synthetic coverage study on a new blind set.
External	**Risk.** The 2019–2025 U.S. primary parameter, availability rules, stable perturbations, and donor radius may not transfer. **Mitigation.** Separate 88502 pipeline, complete event accounting, and parameter-specific reporting.	Results do not establish performance for other pollutants, countries, networks, eras, or perturbation mechanisms.	Independent publicly reproducible benchmarks with precommitted footprint-disjoint evaluation.

10.2 Construct validity

The primary construct is a reported metadata transition paired with a protocol-defined local residual, not a physical monitoring event. Method names may contain useful contextual descriptions, but they do not substitute for technical records. The benchmark therefore cannot determine whether a change reflects hardware, configuration, quality assurance, a reporting convention, or another process. The human-blocked taxonomy remains excluded from all stratified results because a row-by-row interpretation has not been completed by students and supervising teacher.

The v0.5 construct is even narrower: a synthetic local/shared scope label under a target-fixed generator. It is deliberately detached from AQS physical sites and uses no AQS inputs. The partial participation values are comparison design parameters, not measurements of how a regional event propagates. The certificate receives known generator quantities unavailable in a real audit. Accordingly, its result is a test of a structural abstention condition, not external validation of an operational decision system.

10.3 Internal validity

Cross-site counterfactuals require an assumption that qualified donors offer a useful pre-event reference. Physical-site deduplication, correlation, distance, method stability, and paired-data checks reduce obvious violations, but cannot ensure that the target and donor network remain comparable after an anchor. Regional events may be imperfectly shared, and a local residual may have non-instrument causes. Placebos and sensitivity analyses therefore constrain interpretation; they do not identify a unique mechanism.

10.4 Statistical validity

The fixed-weight interval omits uncertainty from donor selection and model selection by construction. The nested interval expands that scope inside a fixed observed candidate pool, yet the evaluation does not include a full selection-aware synthetic coverage result. The observed undercoverage and overcoverage are negative results, not defects repaired by changing confidence levels or block lengths after inspection. Multiple-screening quantities remain exploratory and do not transform all audit rows into formal causal hypothesis tests.

The v0.5 frontier is an empirical maximum over a finite predeclared policy set, calculated on one held-out synthetic generator. The component bootstrap accounts for repeated scope arms within a component but does not expand the policy class, prove conditional calibration, or establish an unrestricted population supremum. Confidence policies that answer no cases at strict calibration targets are retained rather than treated as missing values. Zero observed certificate error is likewise an observation conditioned on the bounded contract, not proof of zero future error.

The implementation also centers residuals with the inclusive $t_0 - 180$ through $t_0 - 15$ calibration slice while displaying a $t_0 - 60$ through $t_0 - 1$ pre-anchor comparison, producing a 46-calendar-date overlap. This does not leak post-anchor outcomes into fitting, but it prevents the displayed pre-anchor residuals from serving as an independent validation sample. The frozen evidence is reported as implemented; no post-hoc non-overlap result is substituted for it.

10.5 External validity

The study's archive years, PM2.5 parameter, daily summary, U.S. network, and synthetic perturbation generators limit generalization. The benchmark also requires sufficiently dense, nearby, stable, and correlated donor records; its applicability will be lower in sparse networks. A separate 88502 pipeline demonstrates only limited transfer feasibility, not a broad replication. Future work should pre-register independent sources, station footprints, metrics, and external documentation before testing any newly proposed algorithmic claim.

11 Reproducibility, integrity, and contributions

11.1 One-time v0.5 execution and receipt-verified handoff

The scope-answerability experiment was source-frozen at the following annotated tag and commit:

```
v0.5.0-answerability-freeze
14fd0fee4fb015e6c661299041e35ff704a27286
```

Before allocating an output directory, its executor acquired the following annotated remote claim tag:

```
v0.5.0-answerability-execution-claim
```

The claim binds the source freeze, protocol hash, execution-manifest hash, allowlisted source-bundle hash, and locked runtime. A local attempt record is written before result allocation, so a post-claim setup failure cannot be silently reused as a new execution opportunity.

The sole authorized execution wrote immutable result files and a receipt with SHA-256 prefix `954fc9b56a8f5266...`. The read-only result verifier checks provenance, tag bindings, source hashes, runtime lock, schemas, full grid accounting, target and $q = 0$ identity, calibration separation, certificate envelope accounting, and deterministic replay. It passed all eight checks after execution. The frozen-result manifest records every artifact's path, bytes, SHA-256, row count, and receipt binding; presentation assets are copied only after their source result files and figure manifest verify against that receipt.

The v0.5 result directory is intentionally not regenerated, overwritten, retuned, or deleted. A reproduction package must use the preserved outputs and read-only verifiers; running the executor again with `-execute` is prohibited by the one-time claim contract. This restriction applies even if a future reader dislikes the observed comparative frontier, strict-policy abstention, or certificate coverage.

11.2 Frozen v0.3.2 evidence and deployment handoff

The v0.3.2 deployment-audit numerical evidence remains bound to the v0.3.2-evidence-final release and commit `57d678ecabebff724d898abe626c9ef80538775b`. The tracked evidence summary names every required scientific artifact, records its SHA-256, and specifies the `stable_full_v2` case-manifest checksum. The paper asset generator refuses a missing, changed, non-v0.3.2, or non-`stable_full_v2` source. It writes every displayed numeric table, vector figure, evidence macro, case-study manifest, and asset checksum into a paper-local manifest. The generator does not run the scientific pipeline or alter the frozen analysis artifacts.

Three presentation-only contracts additionally pin the deterministic stable-window illustration, the implemented inclusive window boundaries, and the QA-collocation evidence ladder. Each names its input SHA-256 values and fails if a display source changes. They make limitations visible in the report without adding any result to, or changing any value in, the frozen evidence release.

Table 20: Frozen evidence release verification record.

Record	Value
Evidence tag	v0.3.2-evidence-final
Frozen evidence commit	`57d678ecabeb`
Synthetic result label	stable_full_v2
Case manifest SHA-256	`065b1b65c231c529...`
Release-gate checks passed	35/35
Document-consistency checks passed	12/12
Markdown number checks passed	57/57
Two-environment core hashes matched	34

Note. The two-environment statement applies only to the listed designated core artifacts in the frozen reproducibility comparison, not to every local file.

The display-only representative-case renderer adds a second provenance layer. It checks the case-selection configuration, geographic-donor table, and public archive source manifest before locally reconstructing the two complete cases. It then compares each recomputed fixed-weight MetaShift effect with the reliability-prior penalty against the saved frozen method result and fails on an absolute mismatch above 10^{-9}. This allows visual inspection of the underlying time series without committing raw EPA archives, raw API responses, credentials, virtual environments, or cache directories.

11.3 Build and review procedure

The formal-paper build invokes, in order, read-only v0.5 manuscript-asset generation, frozen v0.3.2 asset generation, claim-ledger validation, source-structure validation, reference validation, asset determinism validation, a clean LaTeX build, a font audit, and page-by-page rendering for visual review. Generated assets are deterministic; timestamps are confined to validation records rather than used as scientific inputs. The final build record reports the exact source commit, PDF SHA-256, page count, overfull-box result, font report, and visual-review result.

Reproduction claims are deliberately scoped. The two-environment comparison means that the 34 designated core artifacts listed in the frozen comparison matched by SHA-256 under the locked requirements file. It does not assert that every local file, renderer output, timestamp, or untracked raw archive is byte-identical across all systems.

11.4 Student contribution and integrity boundary

This report is a technical draft, not a competition submission. Identity, authorship, advisor relationship, student understanding, AI-use disclosure, taxonomy review, signatures, stamps, plagiarism review, and final attestation remain human-only. Required human steps are in `HUMAN_COMPLETION_CHECKLIST.md`; automation cannot complete them or create a taxonomy-stratified claim.

12 Conclusion

MetaShift-Bench establishes a target-fixed evaluation paradigm for selective scope auditing. Its central object is not an unconditional scope classifier or a promoted estimator. It is the finite-policy answerability frontier: at a specified tolerated scope error, how much of a stipulated scope question can an observation channel answer? The v0.5 construction holds the target observation fixed across local and shared worlds, varies donor participation over a complete adverse-condition grid, and treats a lack of structural separation as mandatory abstention.

The frozen result is deliberately mixed. Under matched balanced target laws, target-only scope inference is impossible at all reported tolerances below one half. A comparative channel has no positive-coverage qualifying policy at the three strict tolerances and gains coverage only at $\alpha = .20$. A synthetic-design-assisted structural certificate has zero observed errors where its declared intervals separate, but it abstains on most nonseparated pairs and is not a deployable real-world policy. These findings reject the simpler claim that comparison information or confidence alone always makes scope answerable.

The prior v0.4 endpoint construction remains an exact sanity check, not the main boundary result. The v0.3.2 AQS audit remains a useful deployment record: it documents how often an observational cross-site comparison can be formed, preserves donor-insufficient and input-window abstentions, and limits metadata-based interpretation. Its frozen estimator comparison does not support a general MetaShift superiority claim. Neither the v0.3.2 audit nor v0.5 synthetic labels establishes a real instrument state, physical mechanism, causal measurement bias, or corrected PM2.5 concentration.

The appropriate next scientific step is not to retune the frozen evidence. It is to evaluate a new preregistered, footprint-independent generator or a separately documented naturally occurring scope setting with admissible external records. Any mechanism-level claim remains contingent on human review of station histories, technical documentation, authorship, and the pending Method Code taxonomy. No taxonomy-stratified analysis is reported.

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A Supplementary protocol details

A.1 Anchor-level audit pseudocode

For each primary-parameter anchor $a = (s, p, t_0, m^-, m^+)$, the frozen workflow executes the following pseudocode:

1. validate canonical daily observations and identify a persistent reported Method Code transition;
2. construct a geographic candidate pool at distinct physical sites, retaining at most one POC per site by the pre-event ranking rule;
3. record an explicit donor-insufficient disposition if fewer than three qualified physical donors exist;
4. fit cross-site weights and residual center only on the inclusive $t_0 - 180$ through $t_0 - 15$ calibration slice;
5. require at least two available donors and at least 30 retained observations in each calibration, pre-anchor, and post-anchor residual window;
6. record an explicit input-window failure if those common conditions do not hold;
7. for complete cases, compute fixed and nested diagnostic intervals, placebo diagnostics, donor sensitivity, and the rule-based evidence tier;
8. preserve every row, including all failures and unavailable diagnostics, in the audit tables.

This sequence is an implementation specification for the frozen audit. It does not claim that any step confirms a physical mechanism.

A.2 Case-study reconstruction contract

The representative-case configuration is a display contract rather than a new experiment. It fixes selection rules before rendering, binds public source archives by their saved SHA-256 values, uses the frozen data-cleaning and reliability-prior weight rule, and records the selected donor physical sites and reconstruction tolerance. If a required archive is unavailable or mismatched, the renderer fails rather than silently substituting a current public download. No raw archive is distributed with this paper.

A.3 Full perturbation-family matrix

Table 21 contains every frozen method, metric, and paired local/regional perturbation family. It is placed in the appendix for readability; Figure 12 gives a deliberately simplified main-text comparison while preserving the complete matrix here.

Table 21: Perturbation-family-specific held-out synthetic metrics for all methods.

Paired perturbation family	Method	MAE	AUPRC	Macro-F1	Regional FPR
Additive step	Standard SC	0.07788	0.99946	0.93092	0.138
Additive step	MetaShift fixed	0.08373	0.99695	0.87969	0.233
Additive step	MetaShift CV	0.07680	0.99801	0.90161	0.193
Additive step	Nearest-neighbor DiD	0.08680	0.99210	0.93614	0.105
Additive step	Bayesian mean shift	0.22860	0.50497	0.41031	0.890
Additive step	Before-after median	0.22774	0.50497	0.44778	0.808
Additive step	CUSUM	N/A	0.50497	0.43785	0.833
Additive step	Rolling MAD	N/A	0.50497	0.49862	0.553
Additive step	PELT	N/A	0.50497	0.42673	0.858
Proportional step	Standard SC	0.07635	0.85856	0.76319	0.188
Proportional step	MetaShift fixed	0.08016	0.84617	0.78293	0.090
Proportional step	MetaShift CV	0.07457	0.84296	0.78357	0.103
Proportional step	Nearest-neighbor DiD	0.08287	0.82011	0.70938	0.065
Proportional step	Bayesian mean shift	0.23444	0.50497	0.48849	0.350
Proportional step	Before-after median	0.22774	0.50497	0.46944	0.260
Proportional step	CUSUM	N/A	0.50497	0.48467	0.328
Proportional step	Rolling MAD	N/A	0.50497	0.40894	0.108
Proportional step	PELT	N/A	0.50497	0.48326	0.320
Gradual drift	Standard SC	0.08321	0.99710	0.93985	0.110
Gradual drift	MetaShift fixed	0.08551	0.99224	0.89419	0.193
Gradual drift	MetaShift CV	0.07958	0.99521	0.92082	0.153
Gradual drift	Nearest-neighbor DiD	0.08746	0.98570	0.93248	0.085
Gradual drift	Bayesian mean shift	0.24325	0.50497	0.45055	0.800
Gradual drift	Before-after median	0.23075	0.50497	0.47143	0.733
Gradual drift	CUSUM	N/A	0.50497	0.45322	0.793
Gradual drift	Rolling MAD	N/A	0.50497	0.49969	0.475
Gradual drift	PELT	N/A	0.50497	0.49992	0.513
Temporary step	Standard SC	0.16187	0.98113	0.91246	0.108
Temporary step	MetaShift fixed	0.16436	0.96551	0.88871	0.130
Temporary step	MetaShift CV	0.15875	0.96994	0.90372	0.113
Temporary step	Nearest-neighbor DiD	0.18885	0.93733	0.84114	0.065
Temporary step	Bayesian mean shift	0.28058	0.50497	0.49930	0.463
Temporary step	Before-after median	0.32696	0.50497	0.49614	0.588
Temporary step	CUSUM	N/A	0.50497	0.48686	0.660
Temporary step	Rolling MAD	N/A	0.50497	0.45747	0.780

Note. Each family aggregates its target-only local perturbation and its matched target-and-donor regional variant. Consequently, regional FPR is assessed within the same paired family; the rows are not local-only effect estimates. N/A has the same meaning as in Table 9.

Table 21, continued

Paired perturbation family	Method	MAE	AUPRC	Macro-F1	Regional FPR
Temporary step	PELT	N/A	0.50497	0.47750	0.708
Variance increase	Standard SC	N/A	0.58680	0.48825	0.158
Variance increase	MetaShift fixed	N/A	0.57484	0.45372	0.105
Variance increase	MetaShift CV	N/A	0.57314	0.46902	0.113
Variance increase	Nearest-neighbor DiD	N/A	0.56157	0.42775	0.055
Variance increase	Bayesian mean shift	N/A	0.50056	0.45595	0.205
Variance increase	Before-after median	N/A	0.50741	0.42520	0.135
Variance increase	CUSUM	N/A	0.50470	0.48012	0.290
Variance increase	Rolling MAD	N/A	0.50191	0.39371	0.093
Variance increase	PELT	N/A	0.49172	0.45495	0.213

Note. Each family aggregates its target-only local perturbation and its matched target-and-donor regional variant. Consequently, regional FPR is assessed within the same paired family; the rows are not local-only effect estimates. N/A has the same meaning as in Table 9.

Table 22: Appendix-only temporal concentration of reported 88101 metadata anchors.

Old code	New code	Abbreviated reported metadata change	Count
236	636	T640, baseline reported name -> T640, alignment-enabled	234
238	638	T640X, baseline reported name -> T640X, alignment-enabled	120

Note. Of 393 2023 anchors, these two pairs account for 354 (90.1%). The largest date is 2023-07-01 (50), and the largest state-code subtotal is 42 (42). Descriptions abbreviate reported AQS Method Name strings; they do not establish a cause.

A.4 Descriptive concentration of reported anchors in 2023

The frozen inventory has 393 of 563 anchors (69.8%) dated in 2023. Two named reported code-pair transitions, 236 to 636 and 238 to 638, account for 354 of those 393 anchors (90.1%). Their new Method Name strings include "Network Data Alignment enabled." Figure 21 and Table 22 make that concentration visible because it is relevant to interpretation and possible future records requests.

This pattern is descriptive only. A code-pair label, a temporal cluster, or an EPA method-context document does not establish an administrative, software, hardware, policy, calibration, or data-processing cause for any site-level record. The human-blocked taxonomy review remains outside this report's stratified analysis.

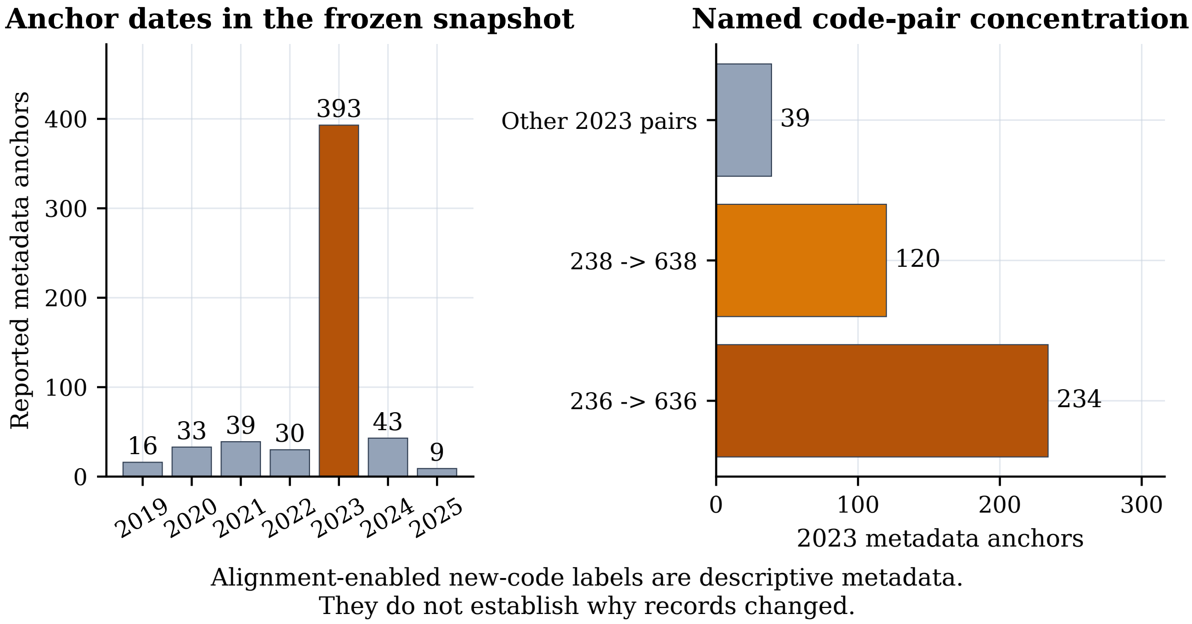

Figure 21: Appendix-only descriptive concentration of reported 88101 metadata anchors. The left panel shows the frozen date distribution; the right panel shows the two named code-pair counts within 2023. The association between labels and dates is not evidence of a physical or administrative cause.

Table 23: v0.5 claim classification and evidence boundary.

Statement	Status	Boundary
Information refinement and reject-option/risk–coverage principles	Established theory	Cited background; not claimed as a MetaShift-Bench invention.
Target-only impossibility and nested-channel monotonicity under the stipulated law	Established theory specialized to v0.5	Requires balanced matched target laws and no target-side leakage.
Availability-normalized $\overline{q_t h_t}$ identity and bounded interval-separation condition	Newly derived construction-specific result	Requires analysis-scale intervention and stated finite envelopes.
Held-out frontiers, gain, risk–coverage, bootstrap interval, and certificate behavior	Frozen empirical observation	Applies only to the finite predeclared policies and generator.
Real AQS scope accuracy, mechanism, sensor diagnosis, or causal physical bias	Explicit nonclaim	Requires independent records, labels, and human review.

A.5 v0.5 answerability protocol and claim classification

The primary v0.5 design is frozen under `v0.5.0-answerability-freeze` and uses a single preauthorized execution. It constructs a complete 640-cell grid for each component; no v0.3.2/v0.4 artifact, external file, taxonomy decision, or AQS record is an input. The following table classifies the claims used in the main text so that the theoretical boundary is not conflated with an empirical observation.

A.6 v0.5 artifact and figure chain

The immutable execution receipt records the exact protocol and execution manifest hashes, source-freeze tag, remote claim tag, runtime lock, row accounting, and output hashes. The raw result bundle contains the pair-level rows, calibration policy, policy metrics, finite-policy frontier, certificate validity, failure map, and component bootstrap. Its figures are rendered only after the result verifier succeeds and carry the receipt hash in their figure manifest. The formal-report asset generator verifies that receipt binding and copies the five source figures byte-for-byte into the manuscript asset directory. It does not rerun the generator or recompute any outcome.

The full post-execution audit and frozen-result manifest retain the result paths, byte counts, SHA-256 values, tag bindings, verifier status, and unfavorable outcomes. These records are intentionally distinct from the v0.3.2 evidence ledger because v0.5 is an independent synthetic experiment with a one-time execution contract.

B Acknowledgements, contribution statement, and required disclosures

HUMAN COMPLETION REQUIRED. This section is an intentionally uncompleted truthful-disclosure template. It must be replaced by a 500–1500 word account, verified by every student author and supervising teacher, before the report is submitted. Do not state that a person performed work they did not perform. Do not state that an advisor, school, organization, or AI system did or did not contribute unless the authors have verified the statement. Before submission, all student authors and supervising teachers must verify the exact final PDF and complete the required academic-integrity declaration, signatures, institutional stamps, and final truthfulness attestation.

Topic origin

HUMAN COMPLETION REQUIRED: Explain how the team selected the question, including the intellectual path from the original observation or reading to the metadata-anchor audit formulation. State whether a supervising teacher or other person suggested the topic and how the students refined the question.

Data acquisition and analysis

HUMAN COMPLETION REQUIRED: Identify who downloaded or reconstructed the public EPA AirData inputs, who inspected data rules, who implemented or reviewed the cleaning and donor-uniqueness logic, and who ran or checked the frozen analyses. State only the actual use of public AQS bulk files, public API endpoints, and local credentials. Do not include credentials, raw API responses, or private information.

Experimental design and writing

HUMAN COMPLETION REQUIRED: State which student(s) designed the synthetic evaluation, maintained the no-tuning boundary, generated figures and tables, interpreted limitations, and drafted or revised each report section. Describe the physical-donor uniqueness defect and remediation only if the authors personally understand and verify that account.

Advisor relationship, compensation, and external help

HUMAN COMPLETION REQUIRED: List each supervising teacher or advisor, their institution and permitted role, the nature of their guidance, and whether any paid tutoring, commercial training, external code, data, or editorial help was involved. Competition rules prohibit untruthful disclosure and may restrict commercial guidance; use the official form for final confirmation.

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